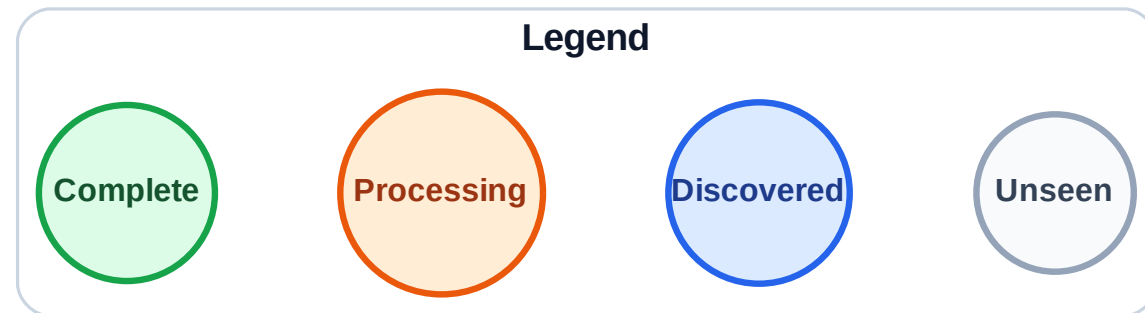
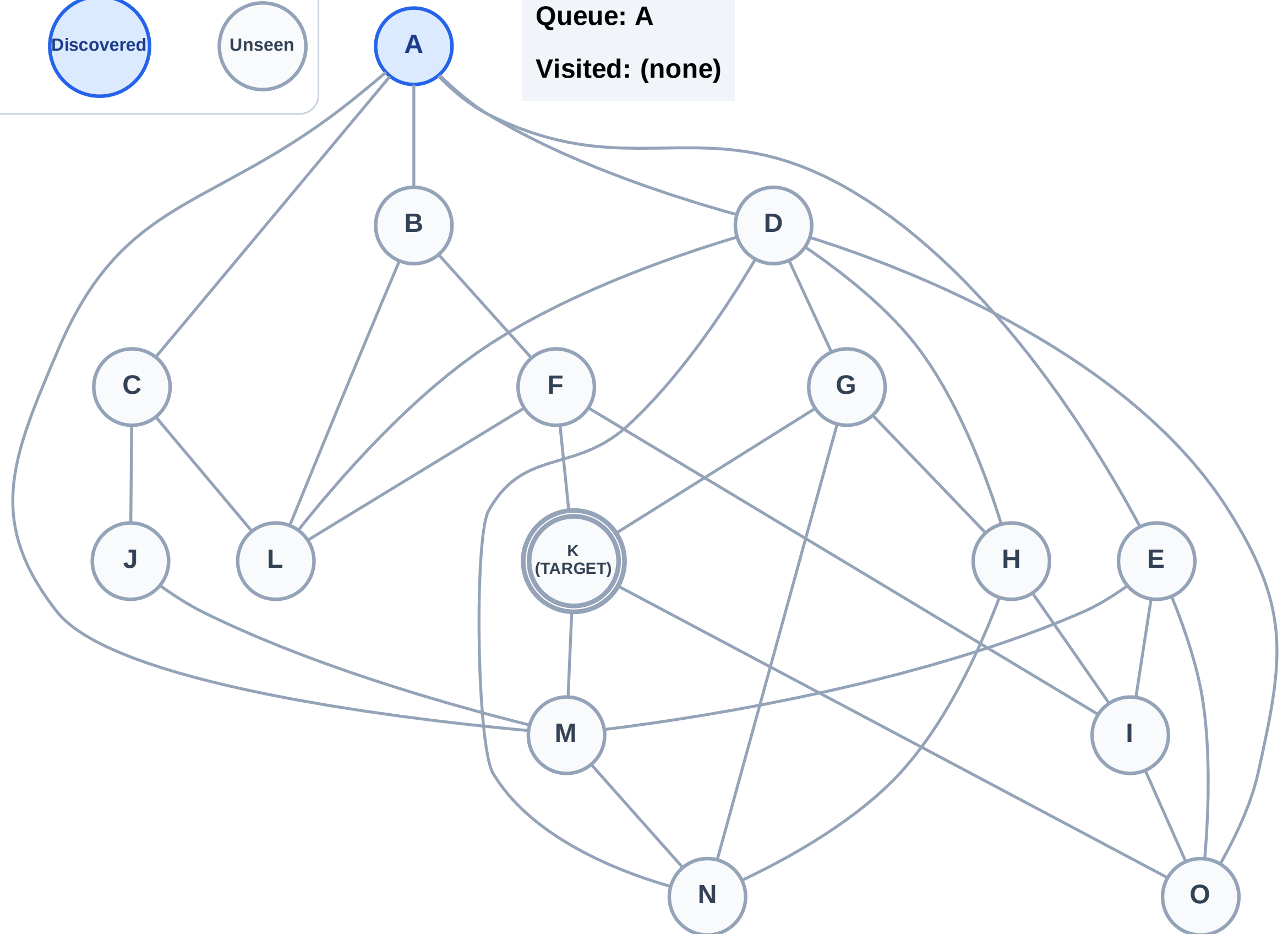


BFS Traversal

Step 1: Start at A (Seeking target node K)



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

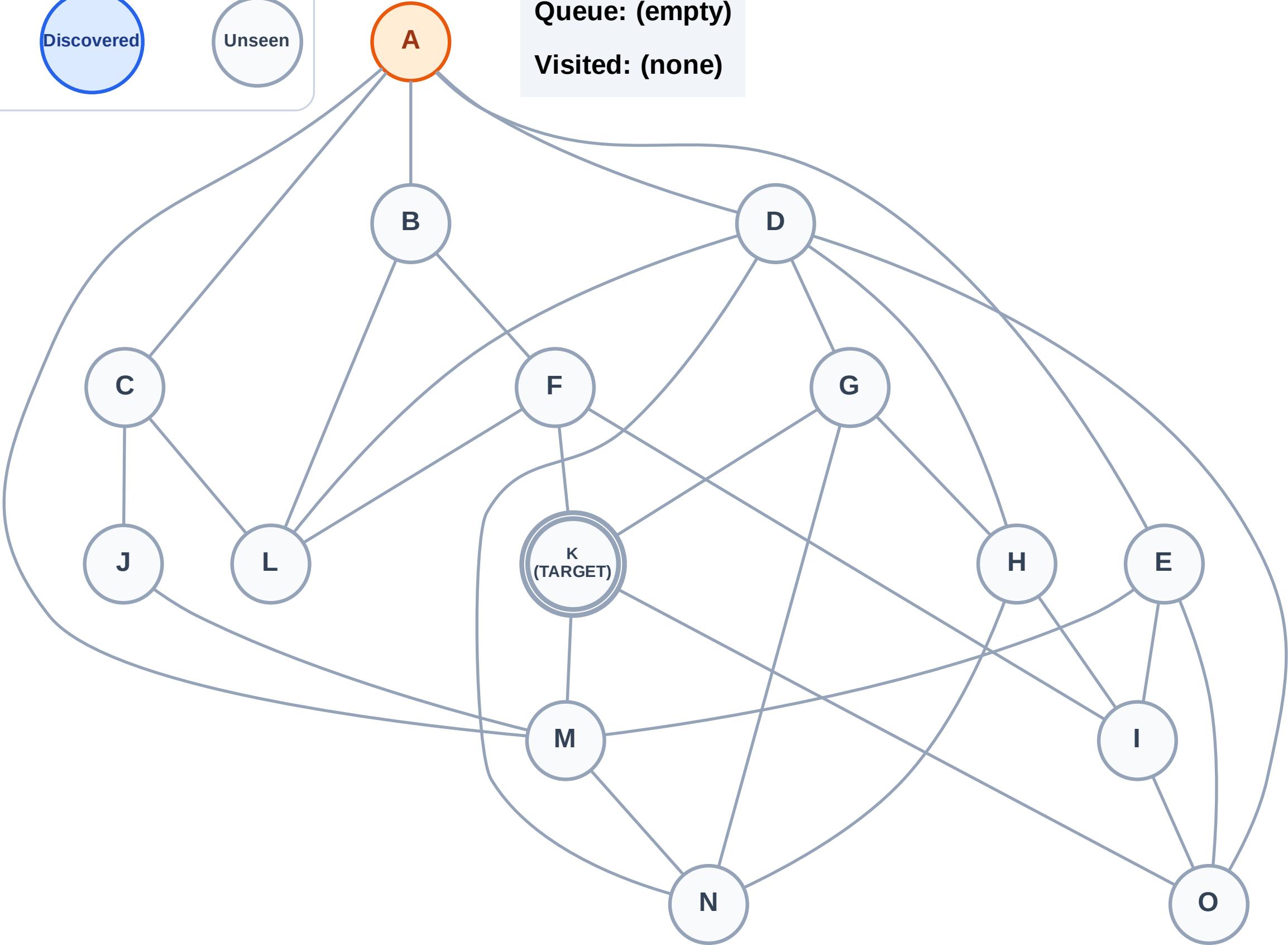
Processing

Discovered

Unseen

Queue: (empty)

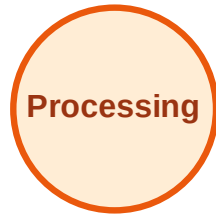
Visited: (none)



BFS Traversal

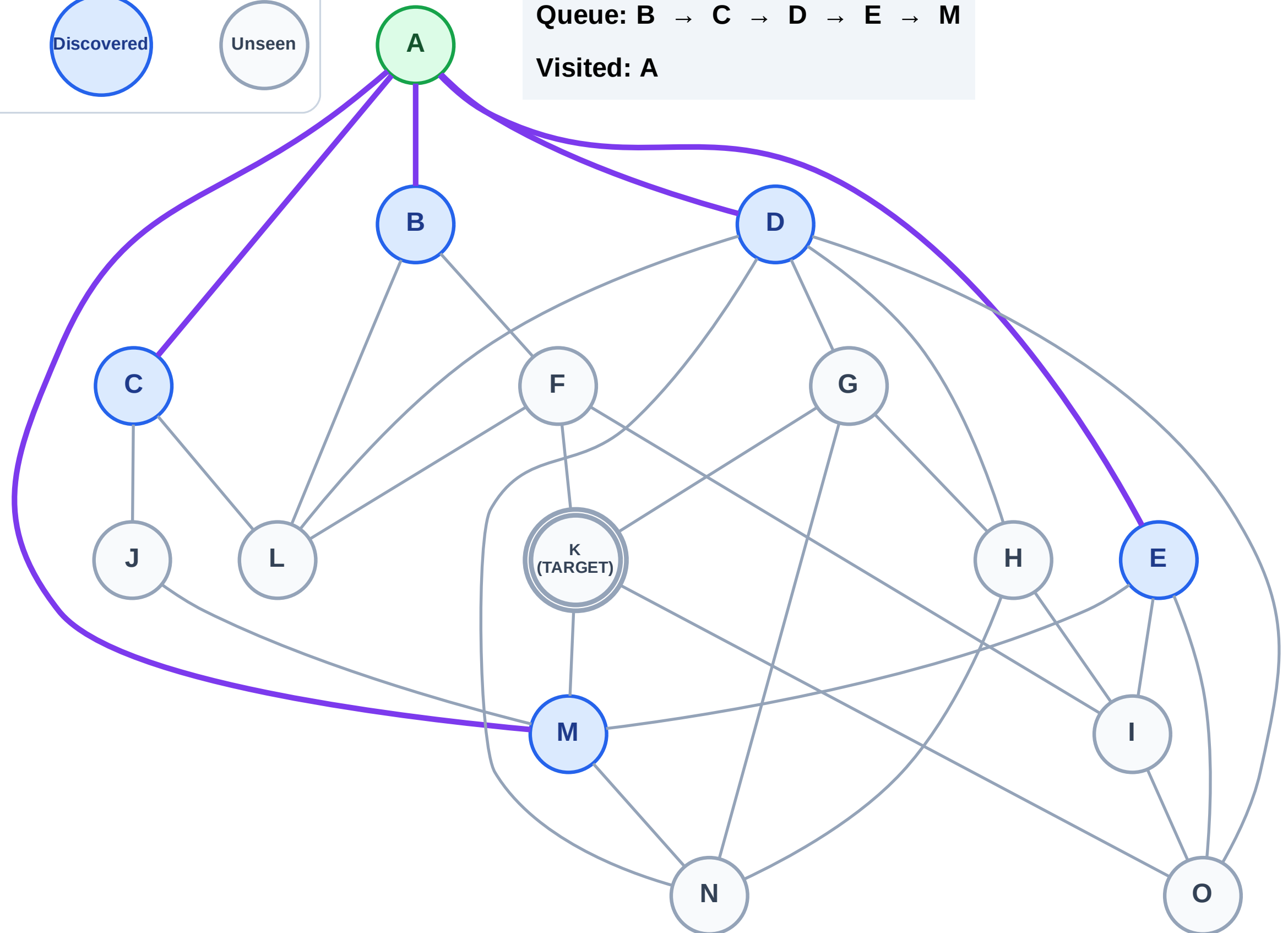
Step 3: Discover B, C, D, E, M from A

Legend

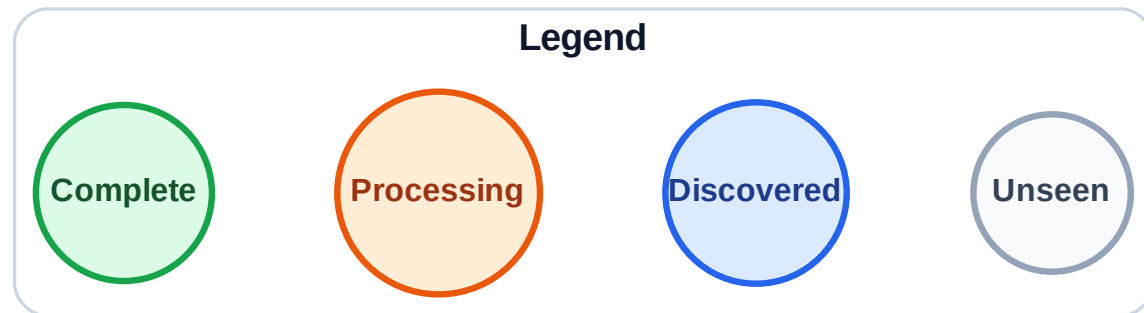


Queue: B → C → D → E → M

Visited: A

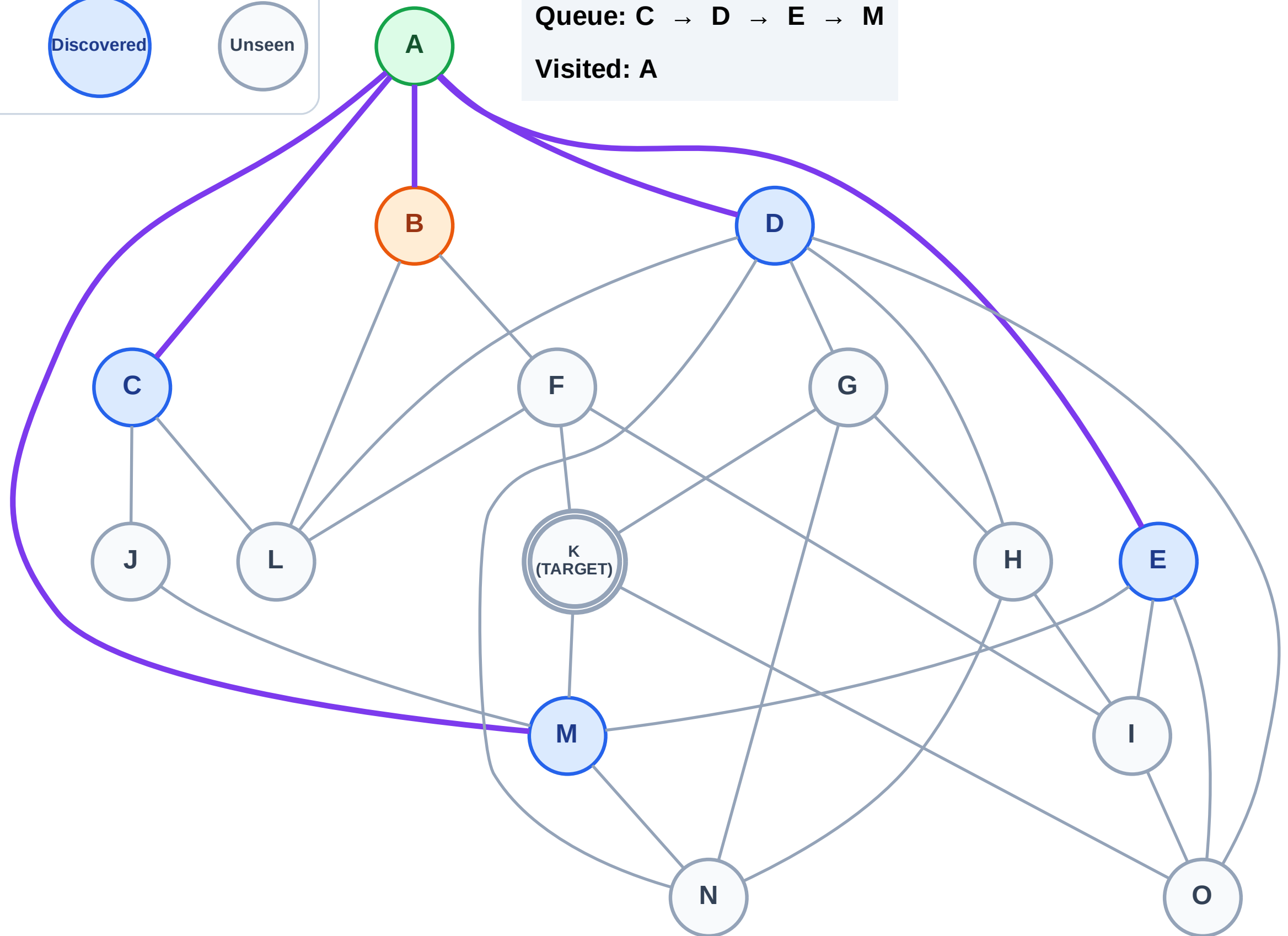


Step 4: Process node B



Queue: C → D → E → M

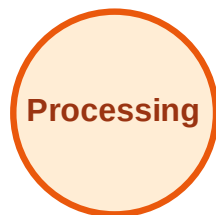
Visited: A



BFS Traversal

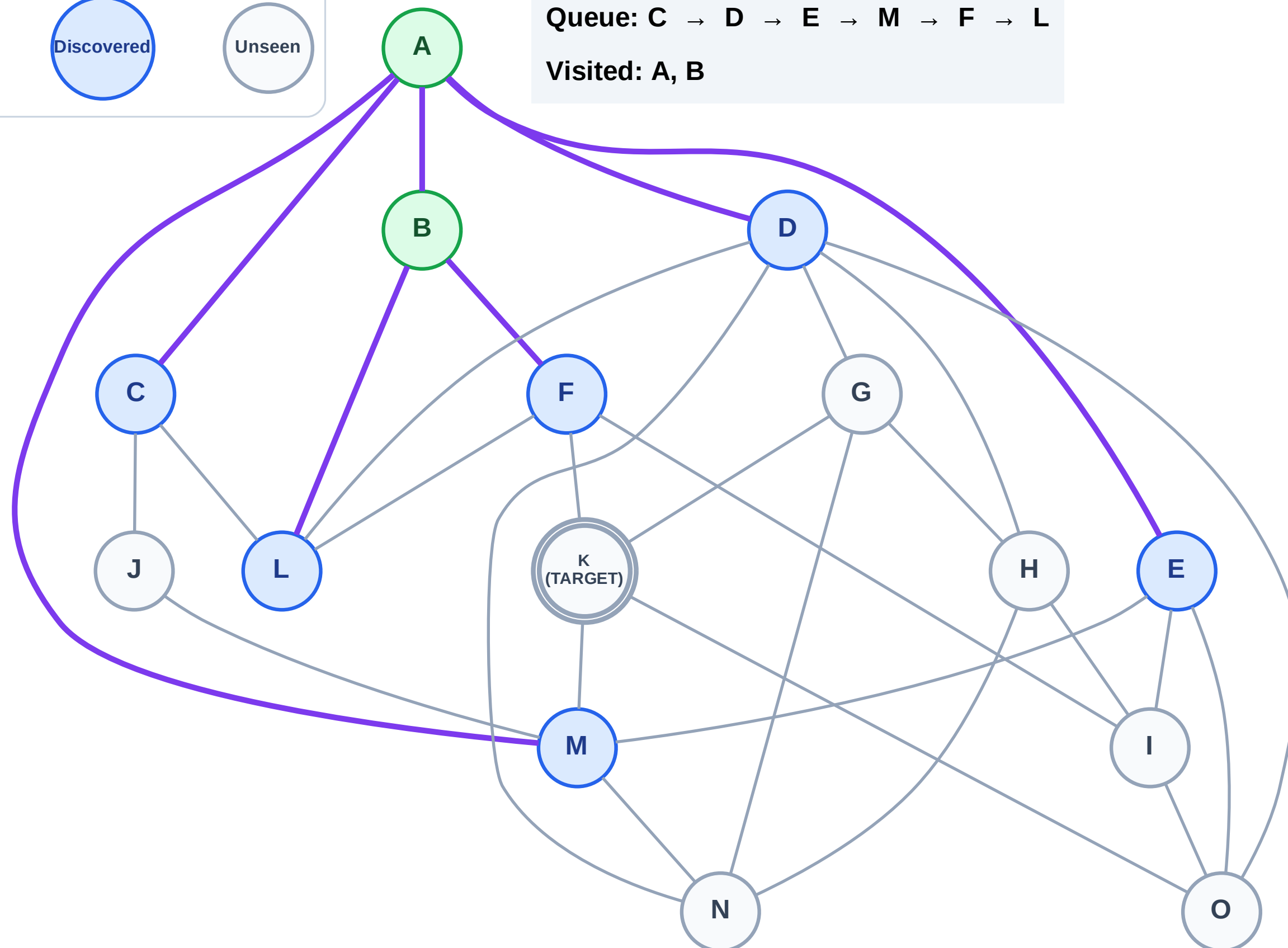
Step 5: Discover F, L from B

Legend



Queue: C → D → E → M → F → L

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

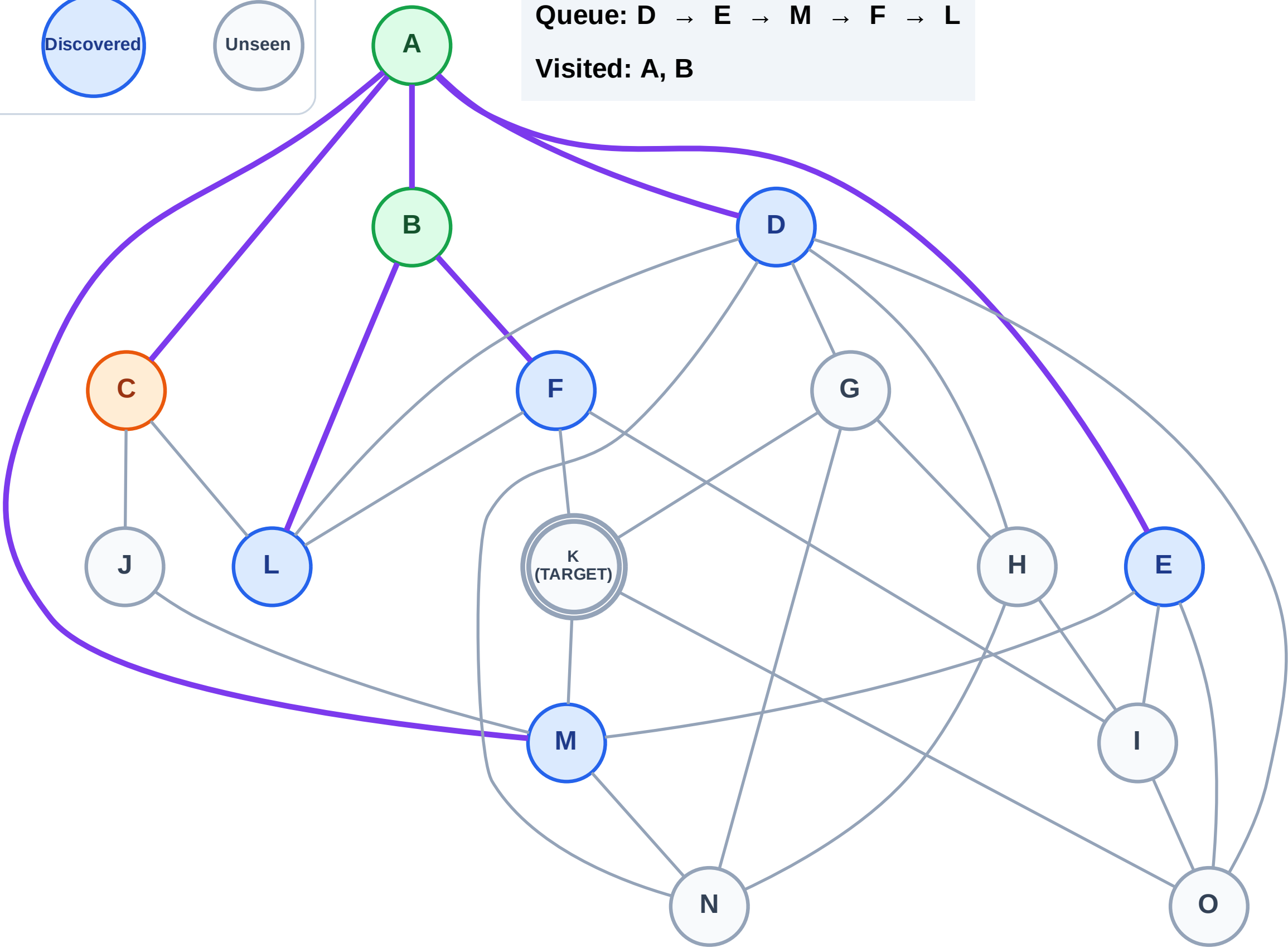
Processing

Discovered

Unseen

Queue: D → E → M → F → L

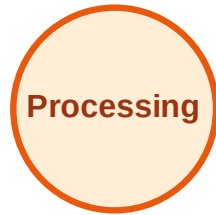
Visited: A, B



BFS Traversal

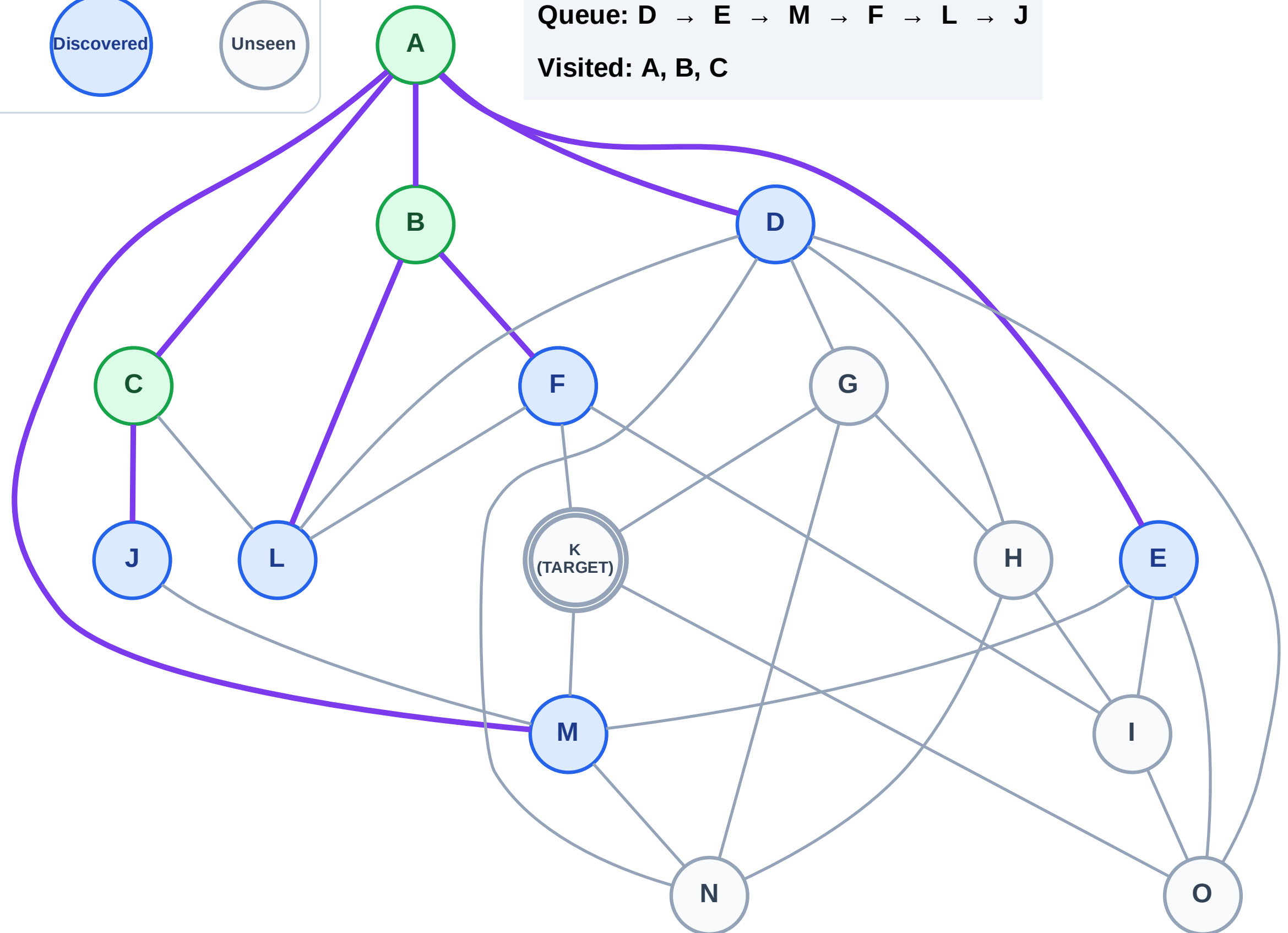
Step 7: Discover J from C

Legend

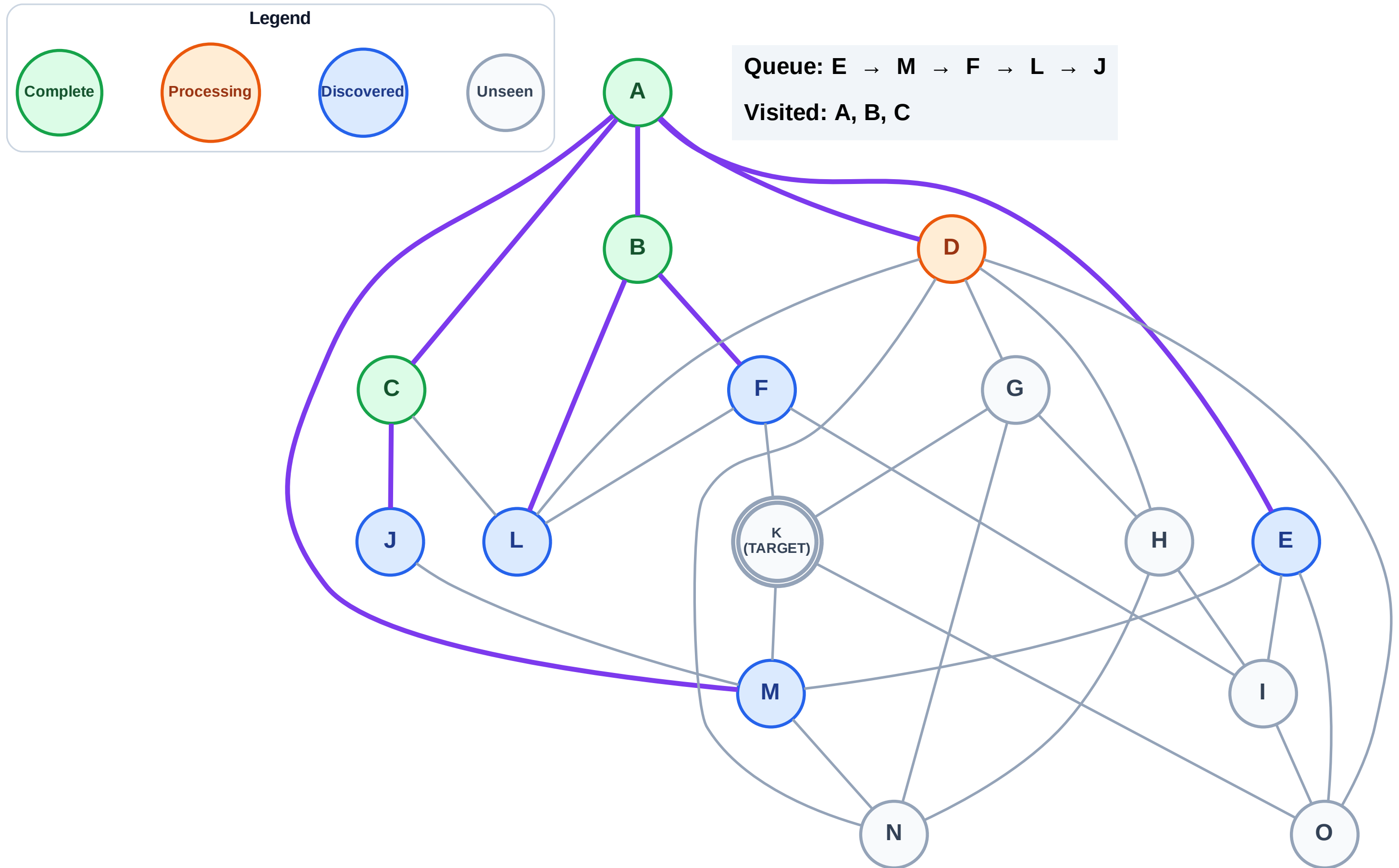


Queue: D → E → M → F → L → J

Visited: A, B, C



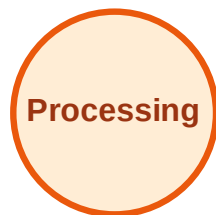
Step 8: Process node D



BFS Traversal

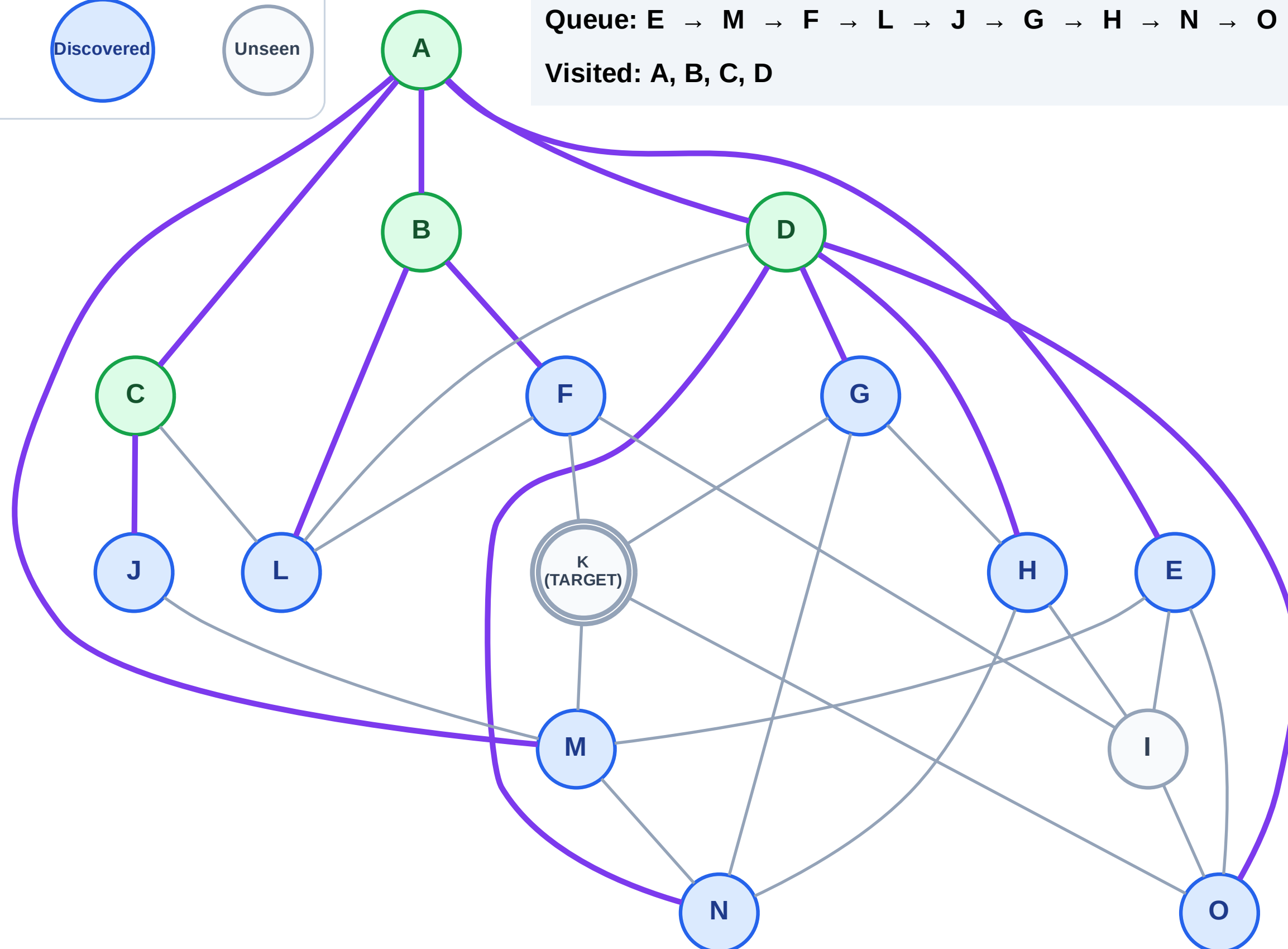
Step 9: Discover G, H, N, O from D

Legend



Queue: E → M → F → L → J → G → H → N → O

Visited: A, B, C, D



BFS Traversal

Step 10: Process node E

Legend

Complete

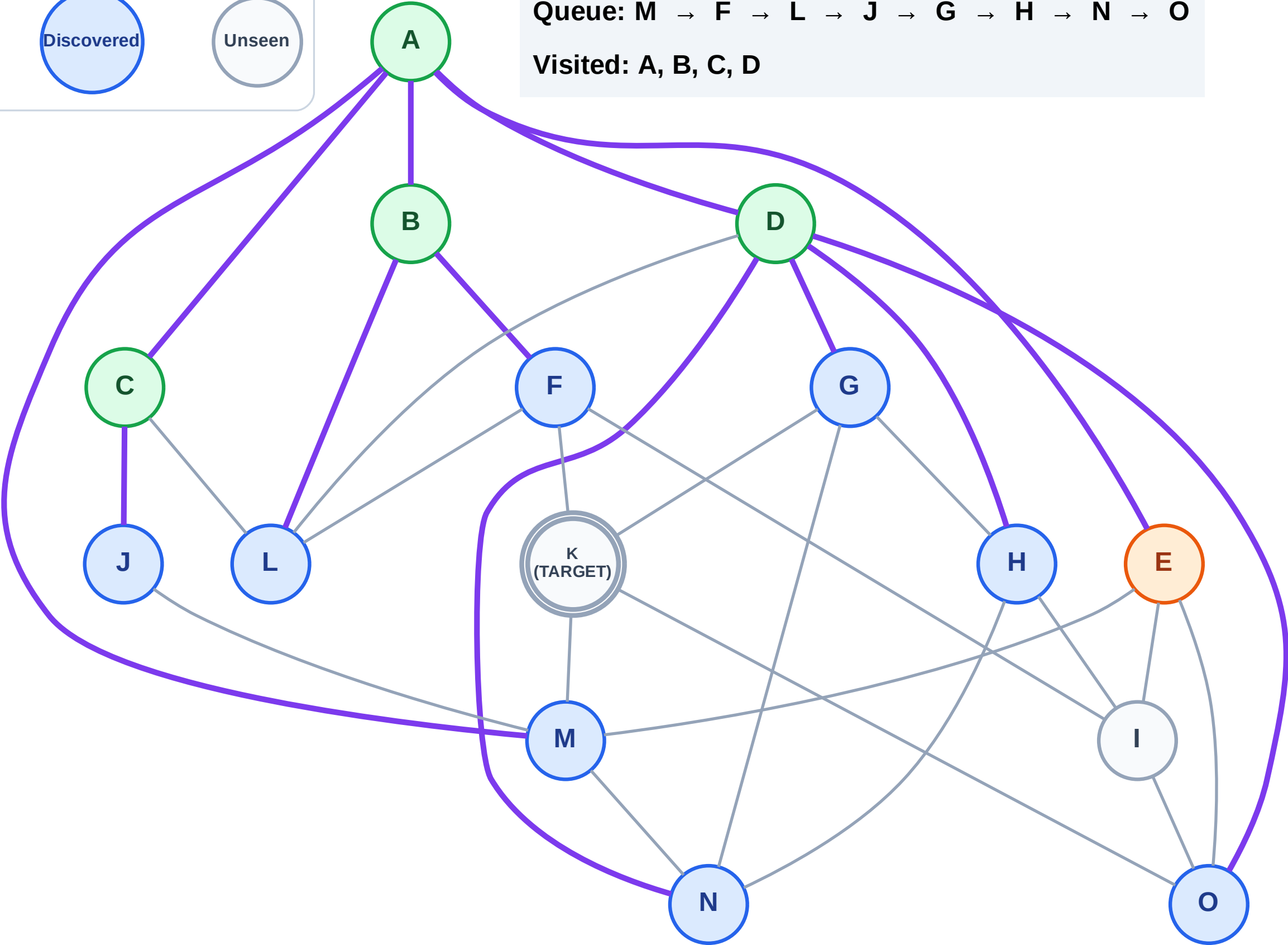
Processing

Discovered

Unseen

Queue: M → F → L → J → G → H → N → O

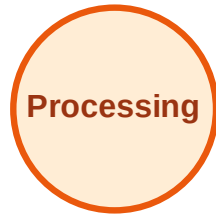
Visited: A, B, C, D



BFS Traversal

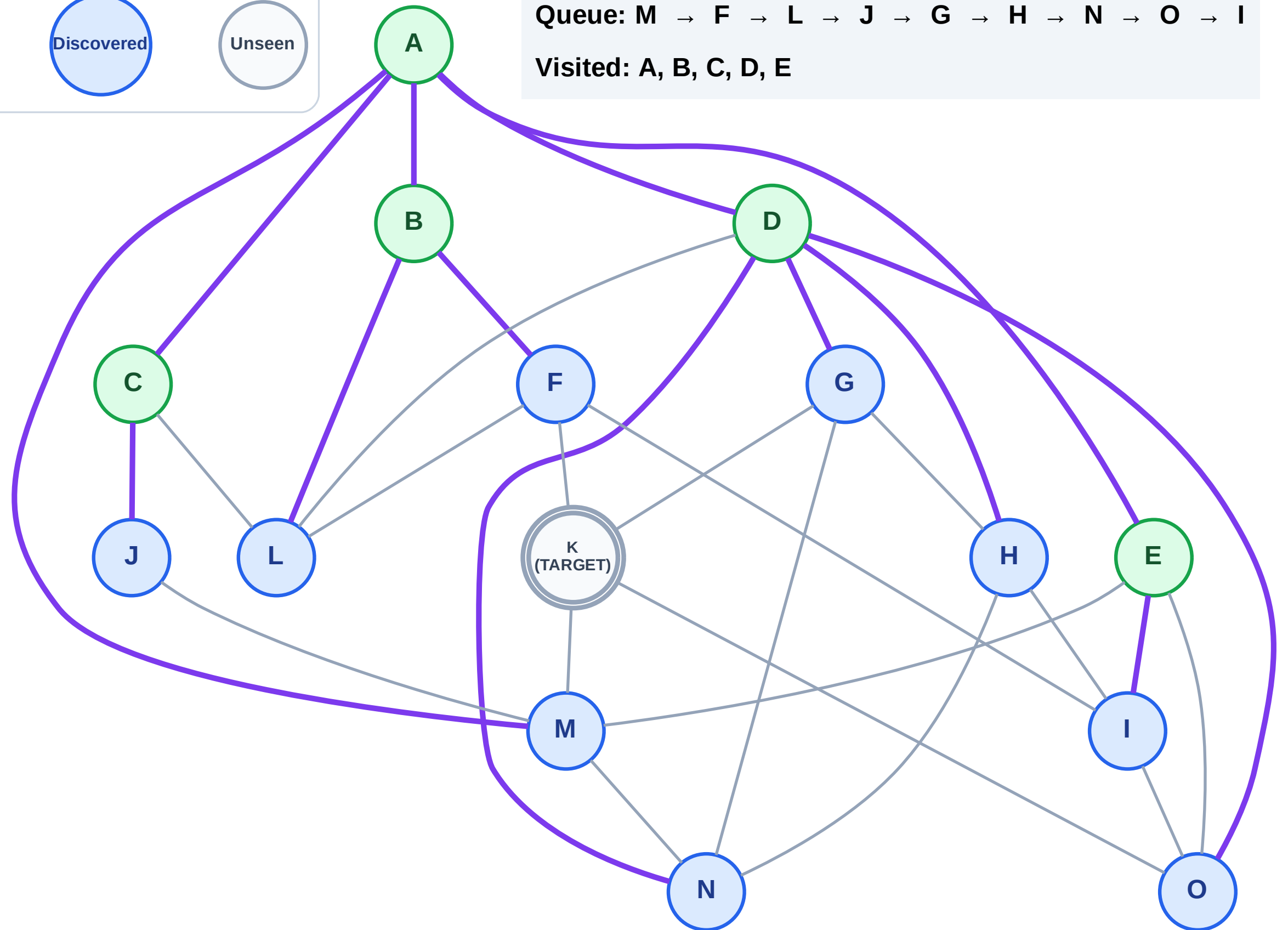
Step 11: Discover I from E

Legend



Queue: M → F → L → J → G → H → N → O → I

Visited: A, B, C, D, E



BFS Traversal

Step 12: Process node M

Legend

Complete

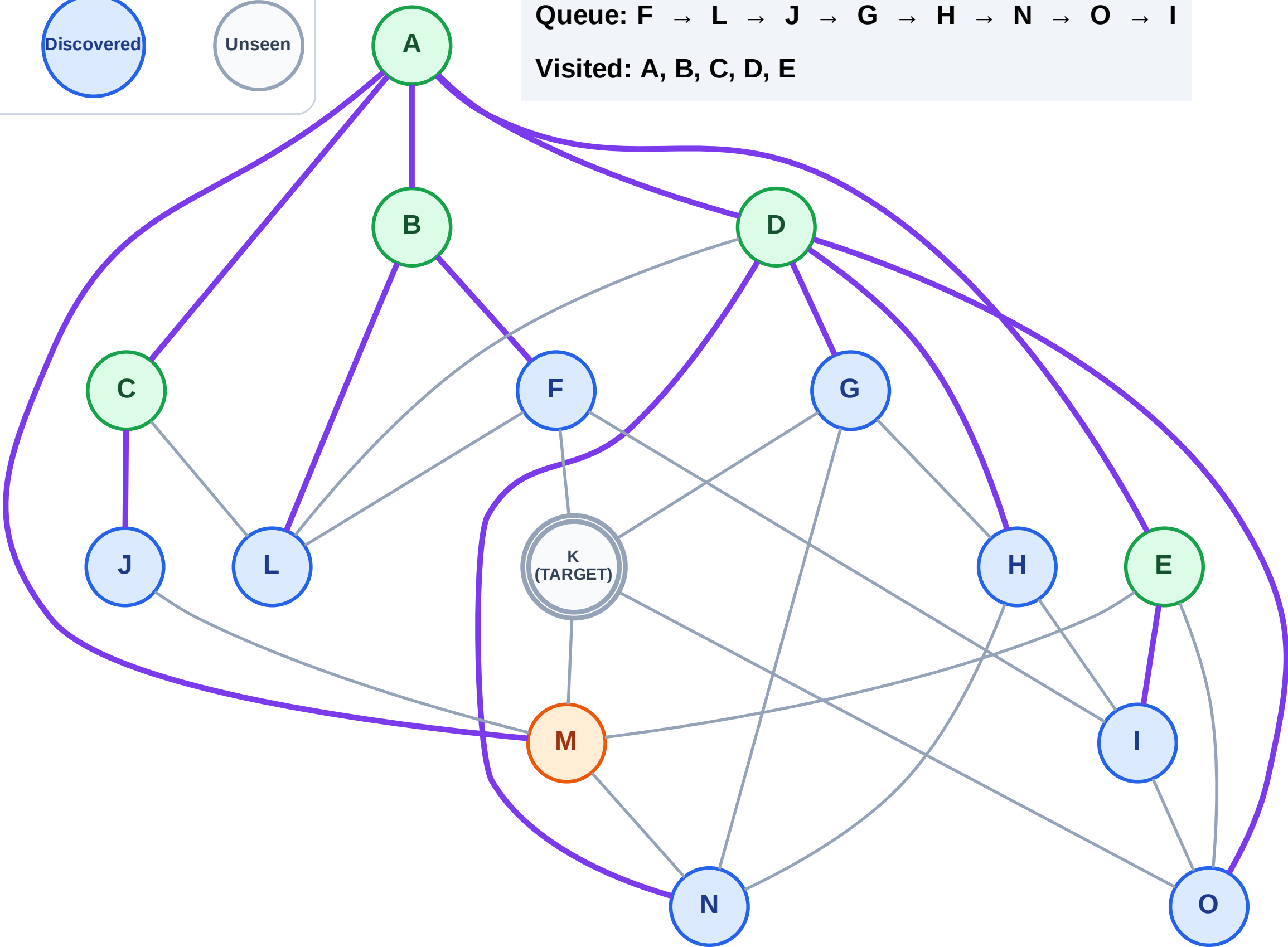
Processing

Discovered

Unseen

Queue: F → L → J → G → H → N → O → I

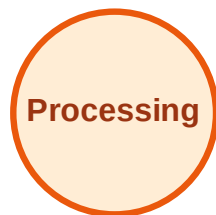
Visited: A, B, C, D, E



BFS Traversal

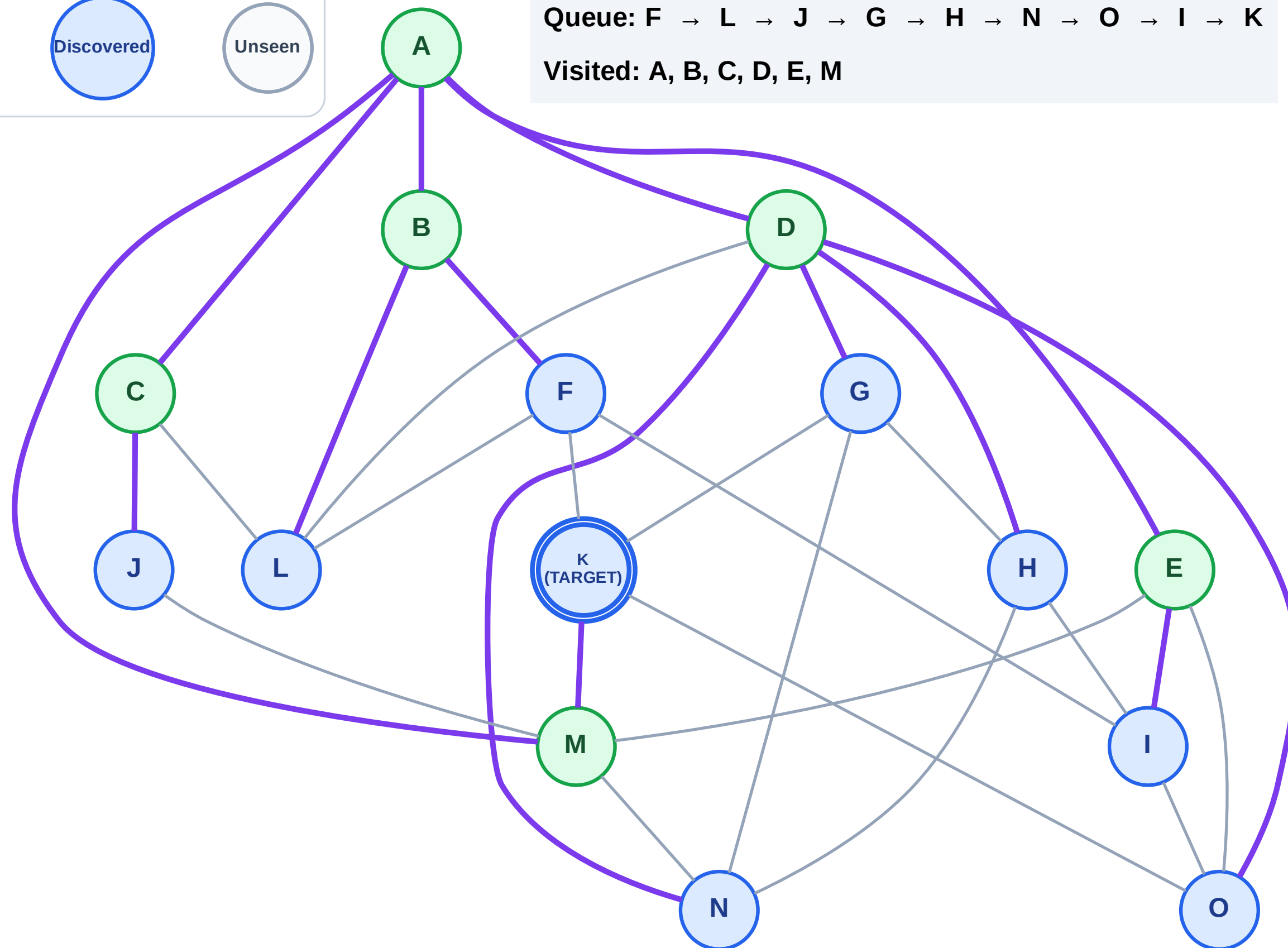
Step 13: Discover K from M

Legend



Queue: F → L → J → G → H → N → O → I → K

Visited: A, B, C, D, E, M



BFS Traversal

Step 14: Process node F

Legend

Complete

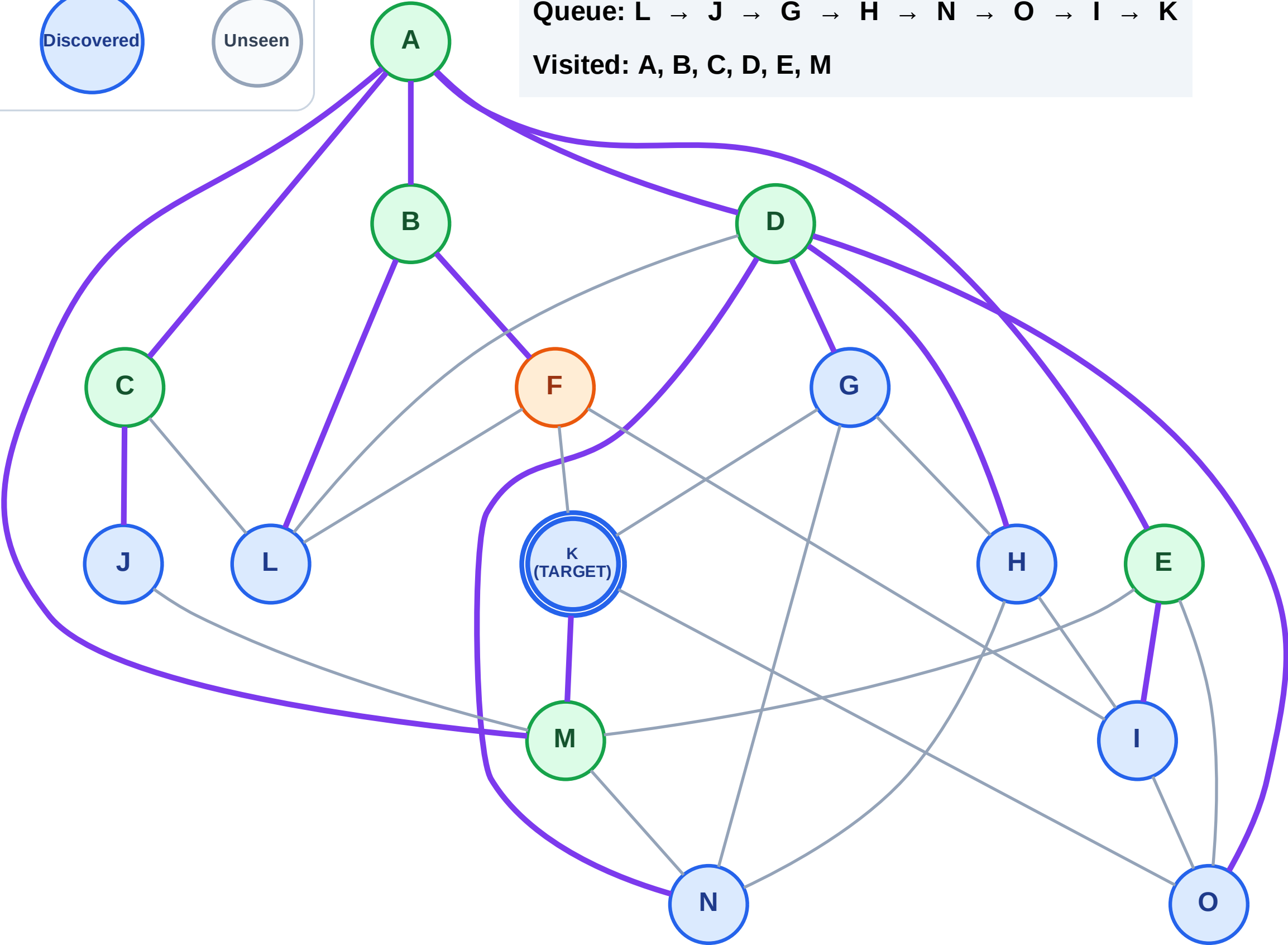
Processing

Discovered

Unseen

Queue: L → J → G → H → N → O → I → K

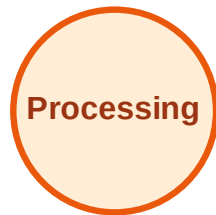
Visited: A, B, C, D, E, M



BFS Traversal

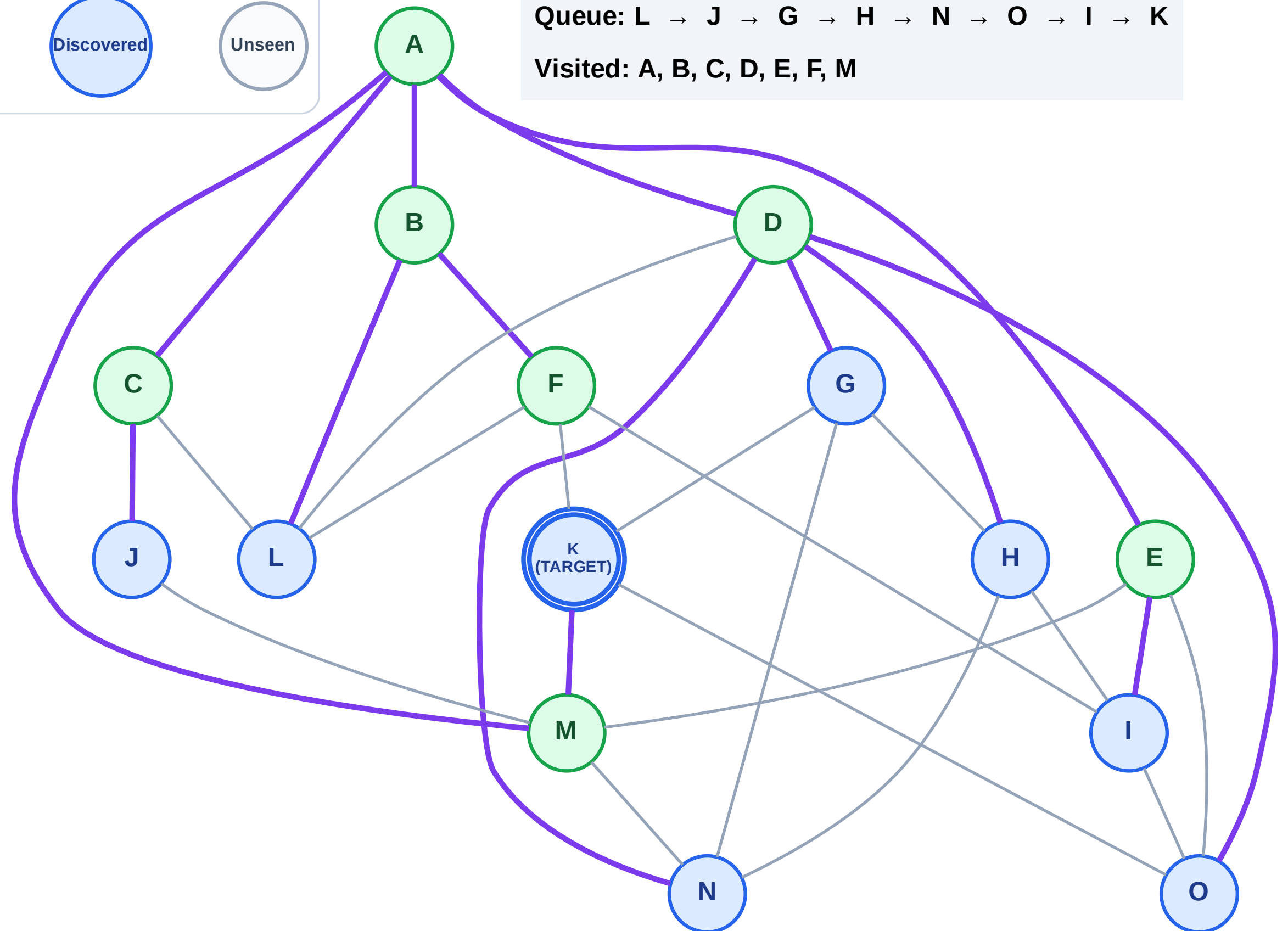
Step 15: No new neighbors from F

Legend



Queue: L → J → G → H → N → O → I → K

Visited: A, B, C, D, E, F, M



BFS Traversal

Step 16: Process node L

Legend

Complete

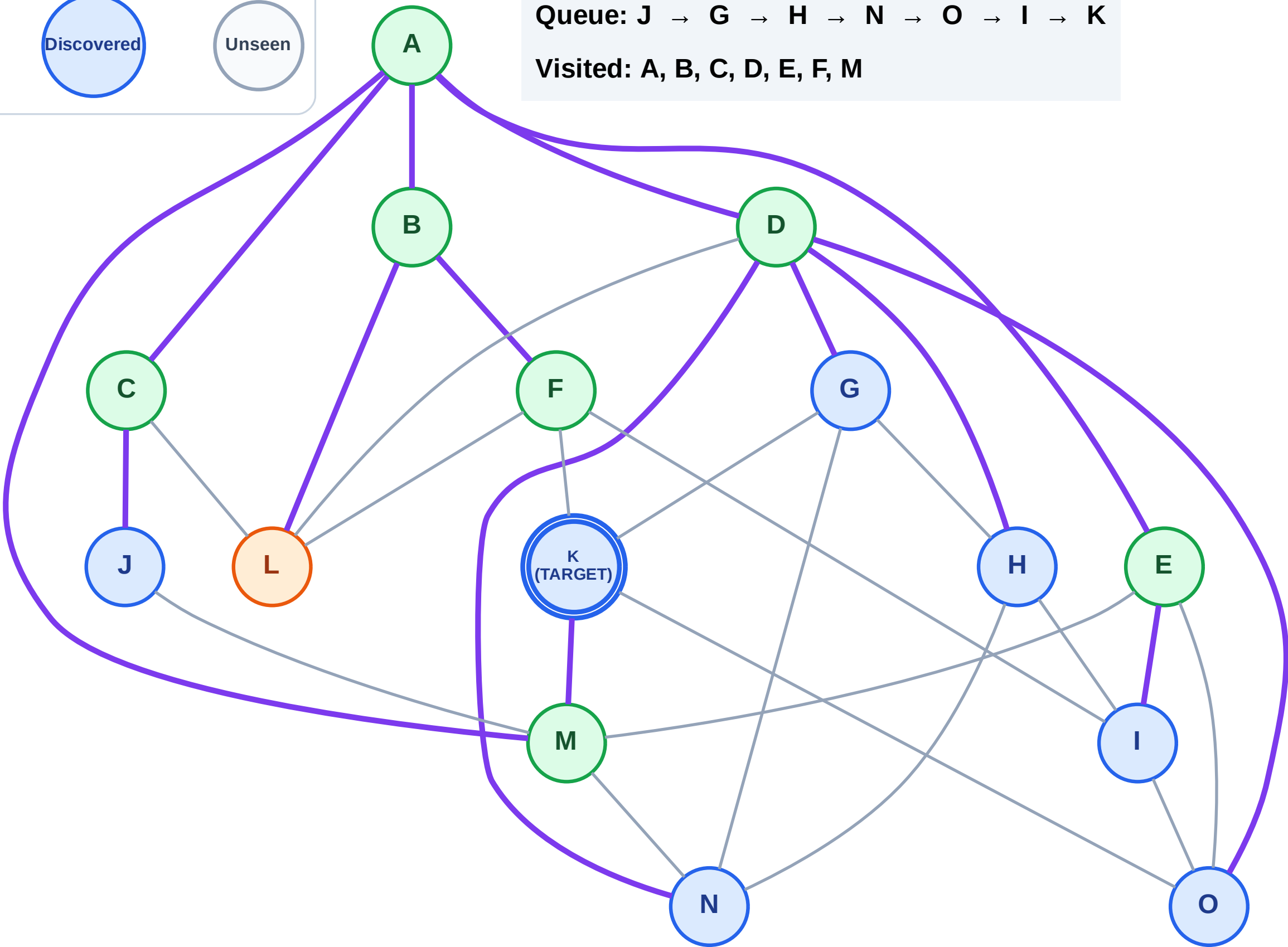
Processing

Discovered

Unseen

Queue: J → G → H → N → O → I → K

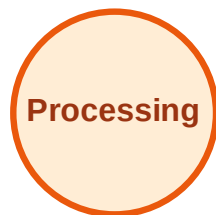
Visited: A, B, C, D, E, F, M



BFS Traversal

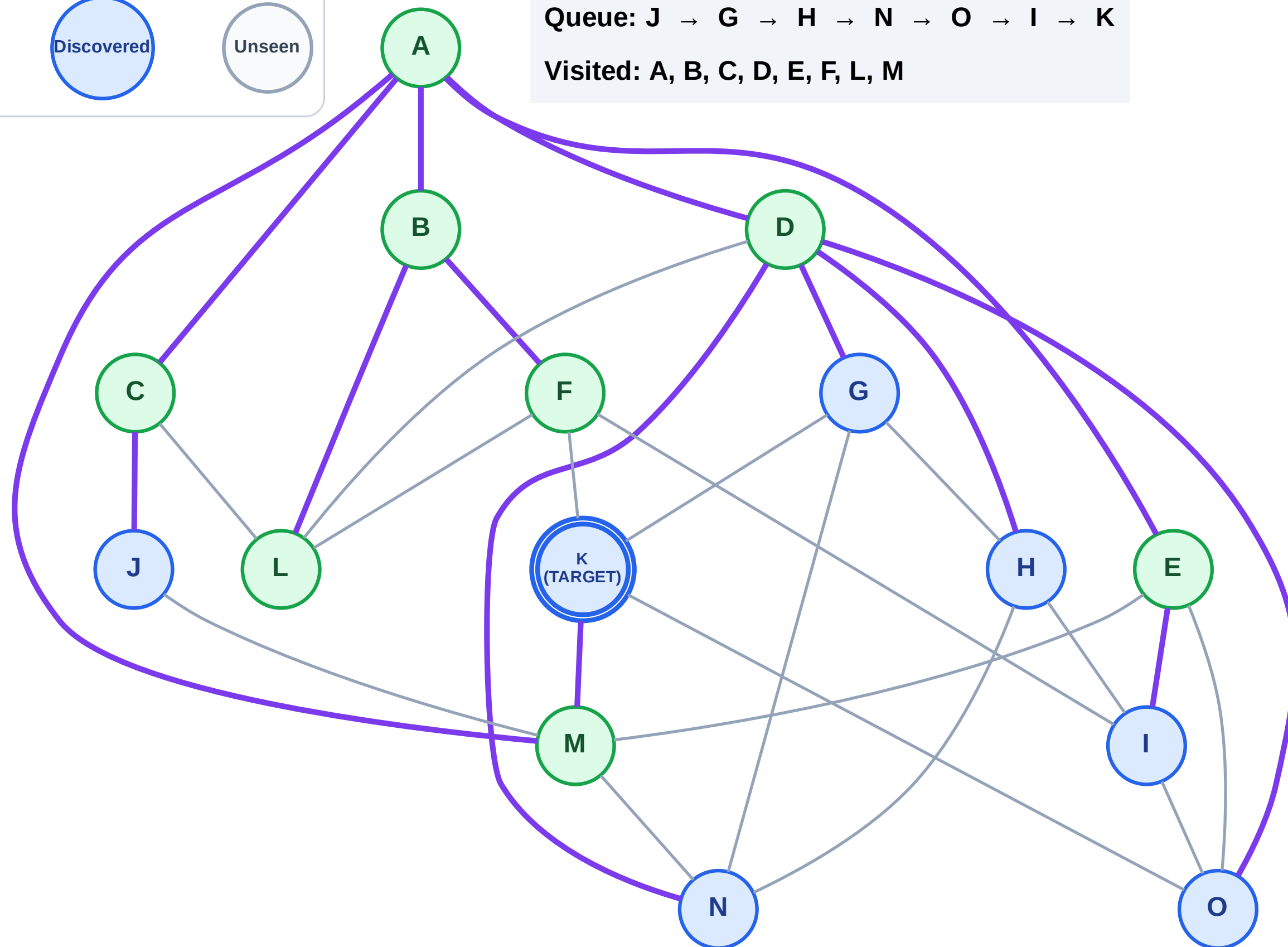
Step 17: No new neighbors from L

Legend



Queue: J → G → H → N → O → I → K

Visited: A, B, C, D, E, F, L, M



BFS Traversal

Step 18: Process node J

Legend

Complete

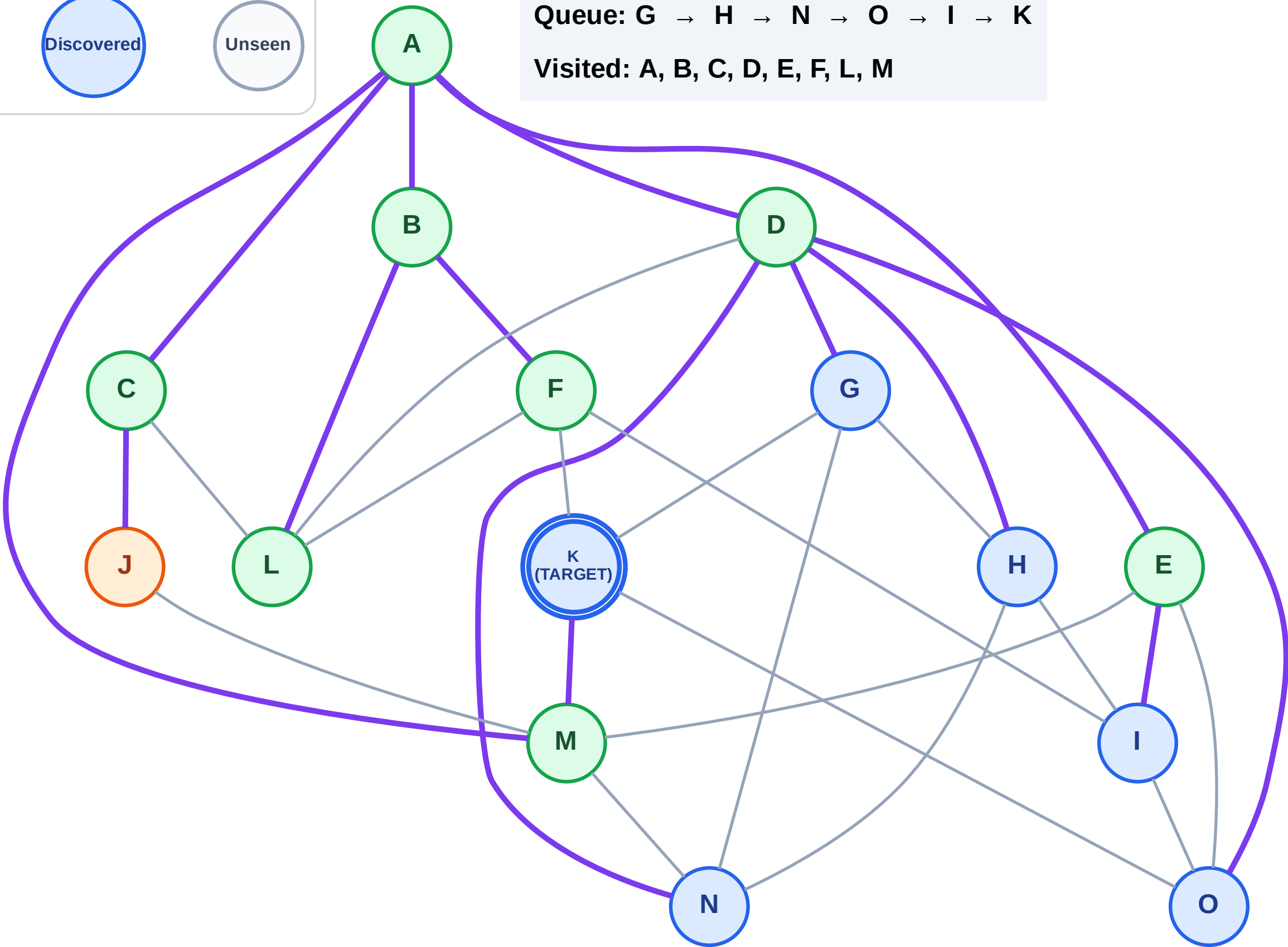
Processing

Discovered

Unseen

Queue: G → H → N → O → I → K

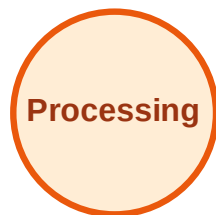
Visited: A, B, C, D, E, F, L, M



BFS Traversal

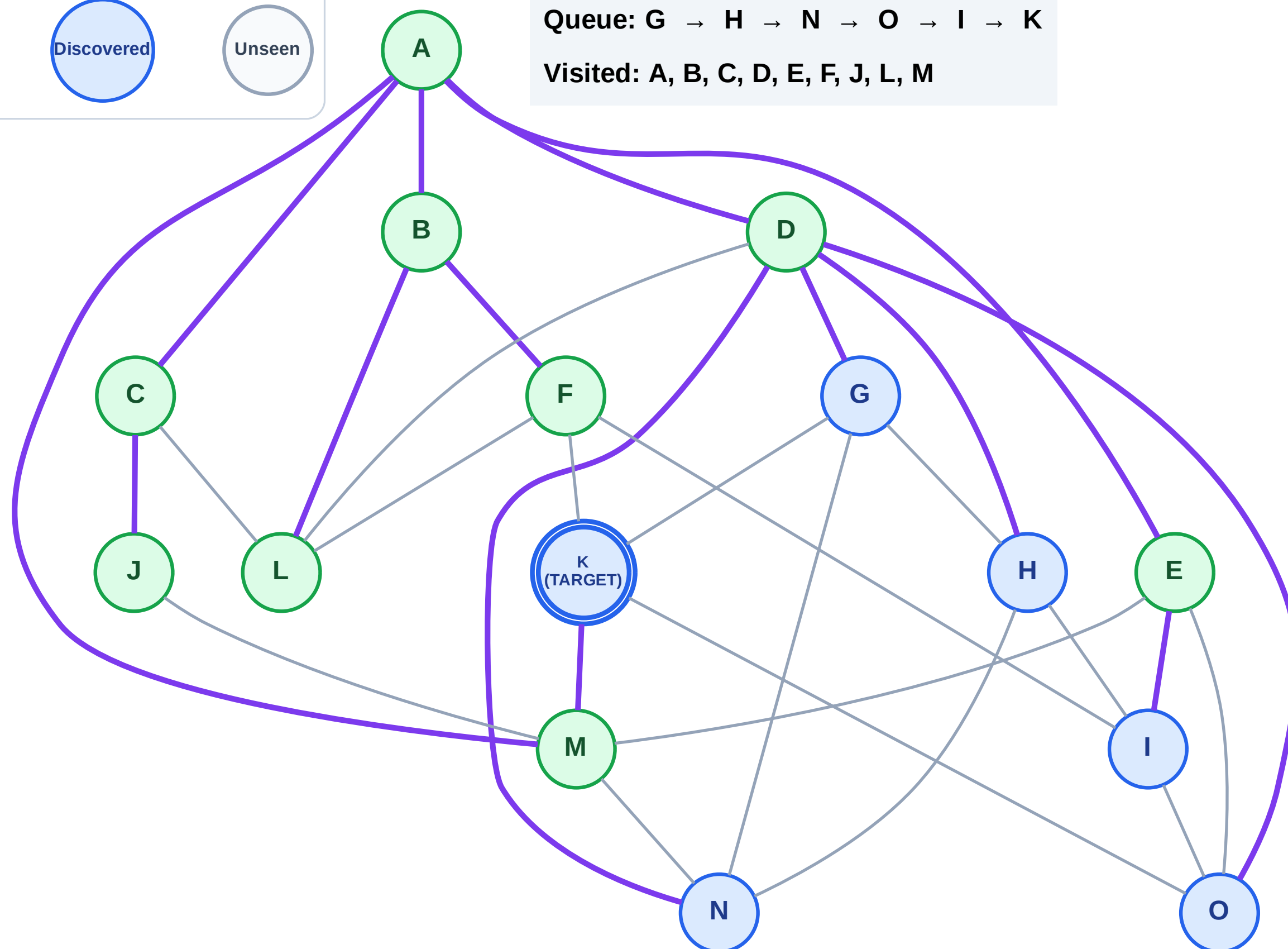
Step 19: No new neighbors from J

Legend



Queue: G → H → N → O → I → K

Visited: A, B, C, D, E, F, J, L, M



BFS Traversal

Step 20: Process node G

Legend

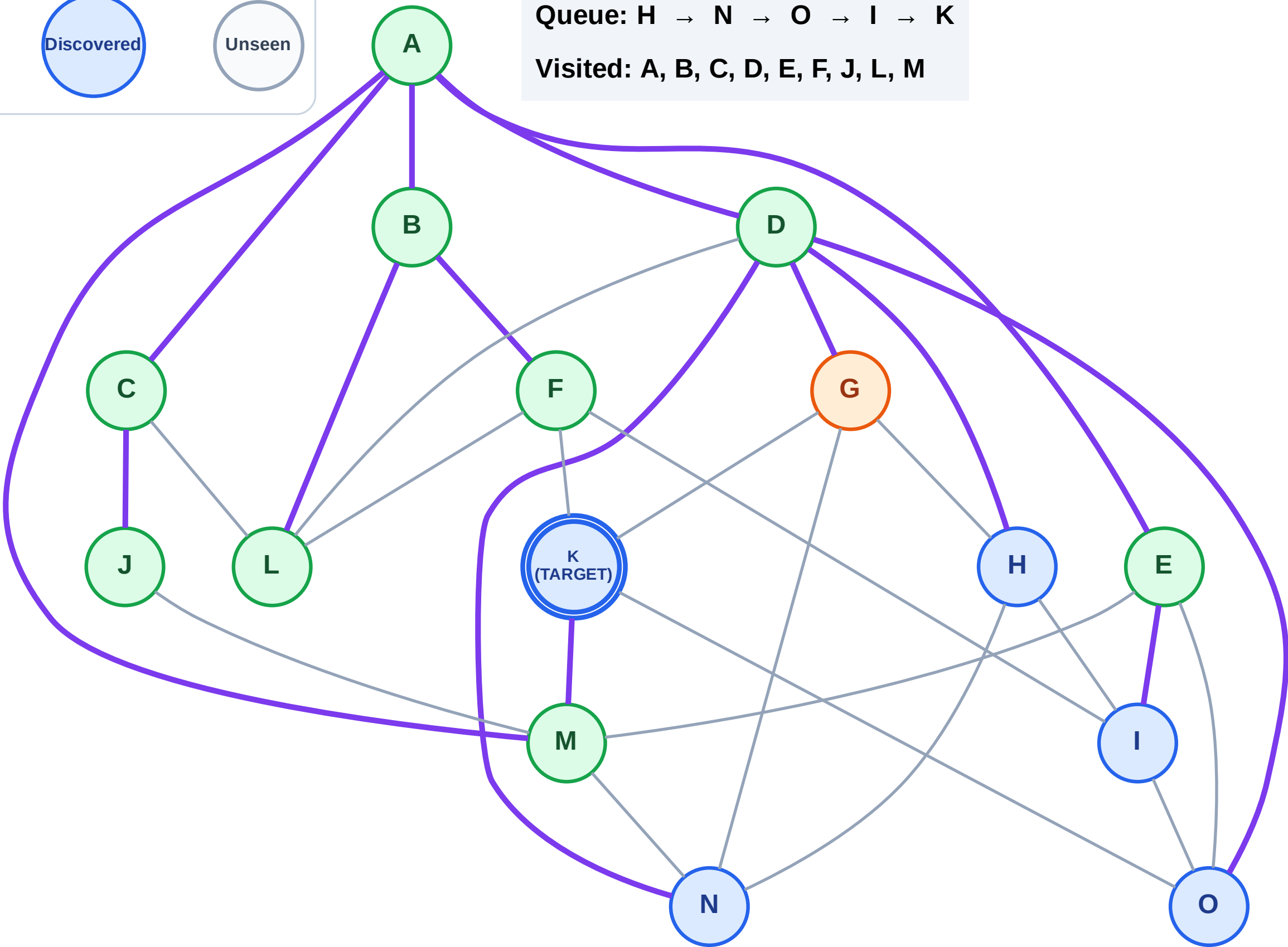
Complete

Processing

Discovered

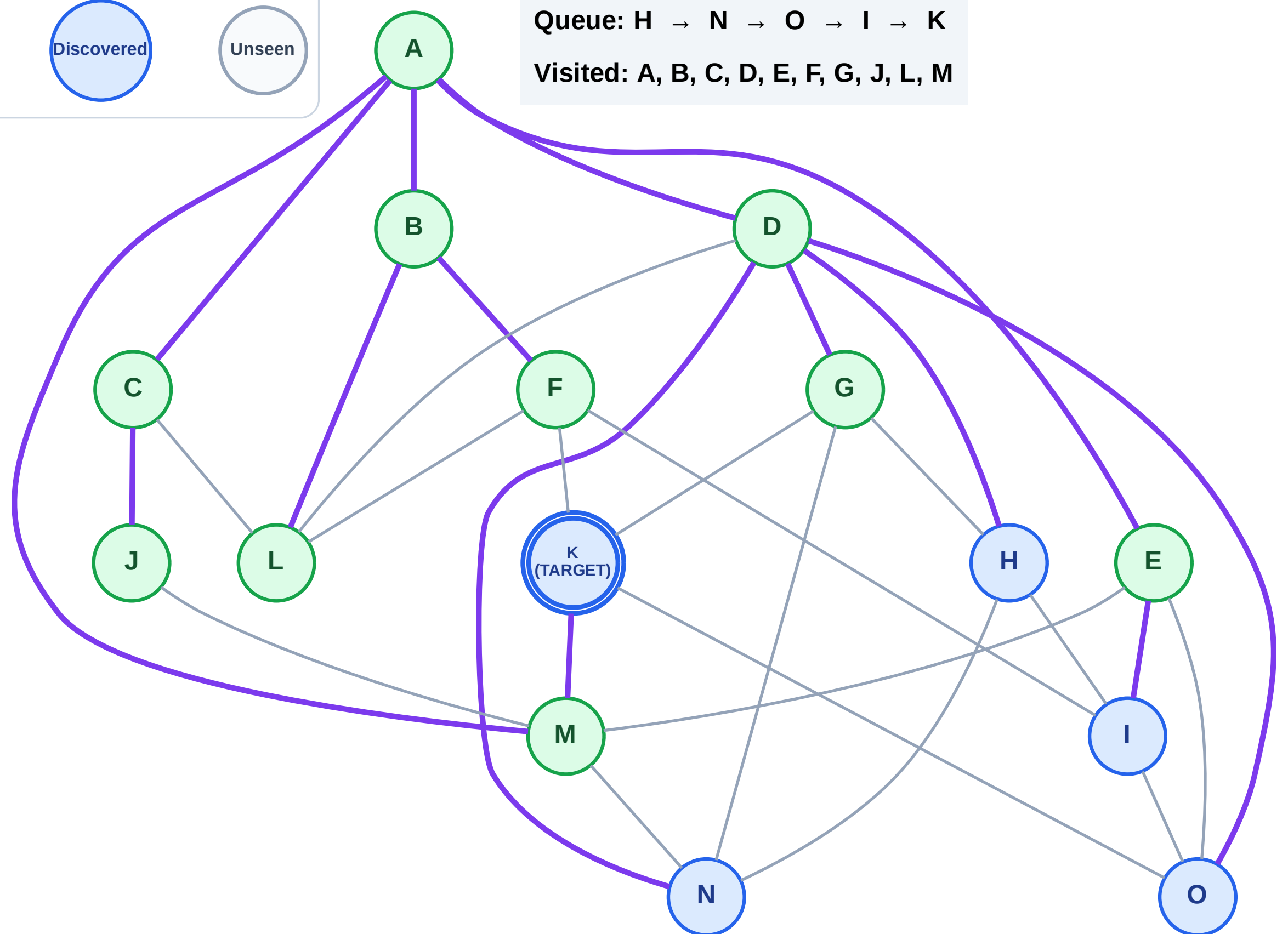
Unseen

Queue: H → N → O → I → K
Visited: A, B, C, D, E, F, J, L, M



Step 21: No new neighbors from G

Queue: H → N → O → I → K
Visited: A, B, C, D, E, F, G, J, L, M



BFS Traversal

Step 22: Process node H

Legend

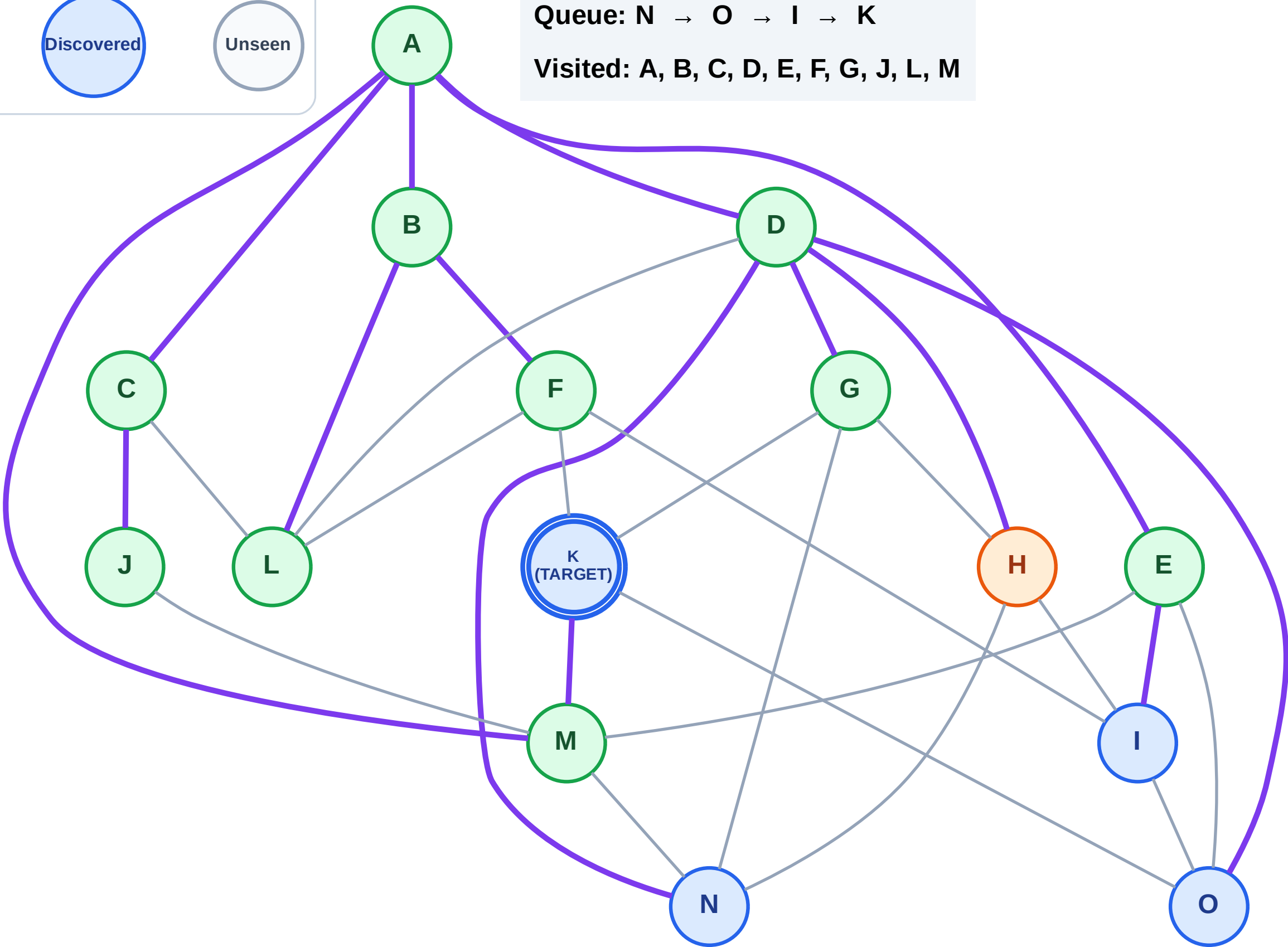
Complete

Processing

Discovered

Unseen

Queue: N → O → I → K
Visited: A, B, C, D, E, F, G, J, L, M

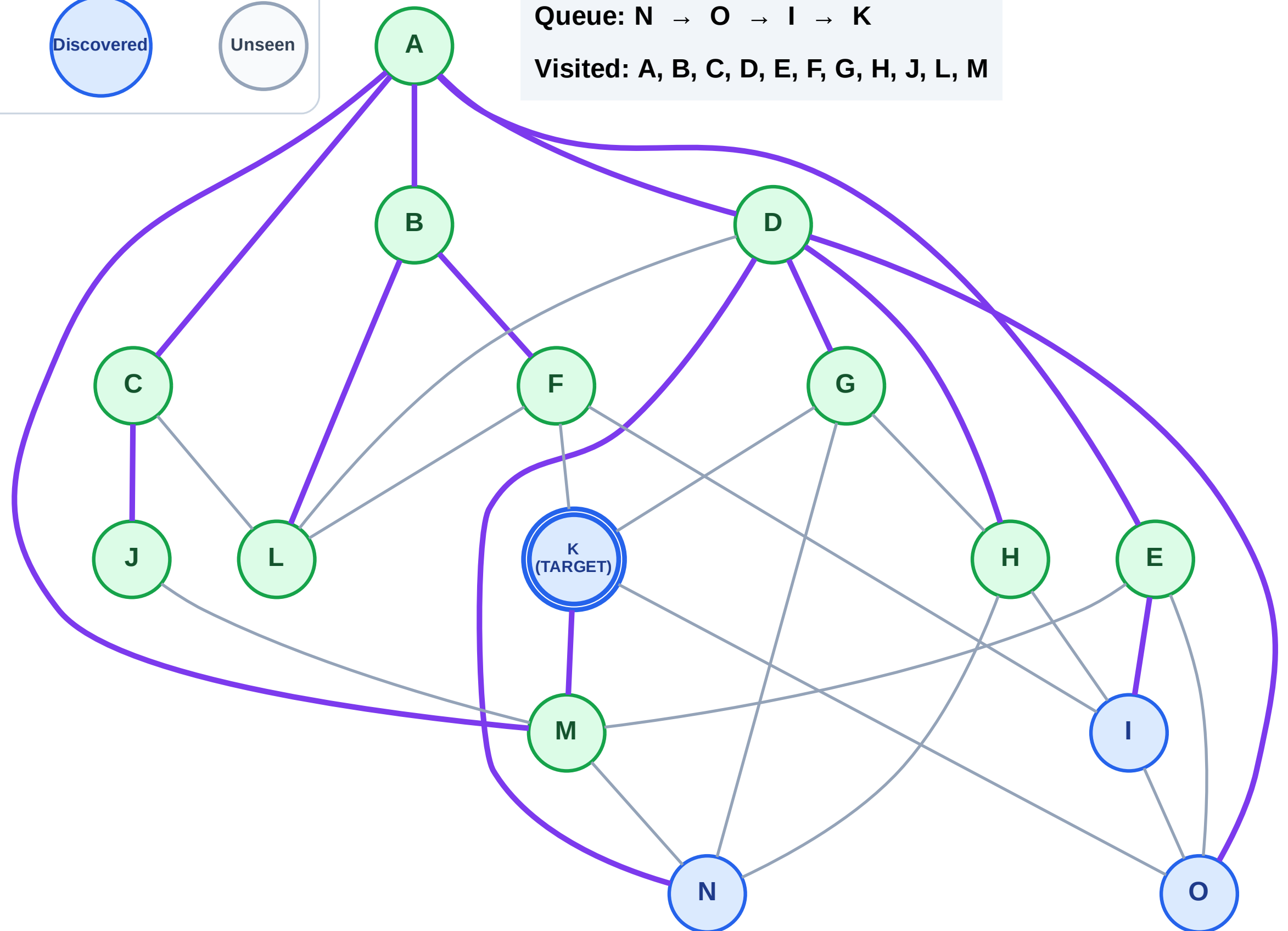


Step 23: No new neighbors from H

Legend

Unseen

Visited: A, B, C, D, E, F, G, H, J, L, M



BFS Traversal

Step 24: Process node N

Legend

Complete

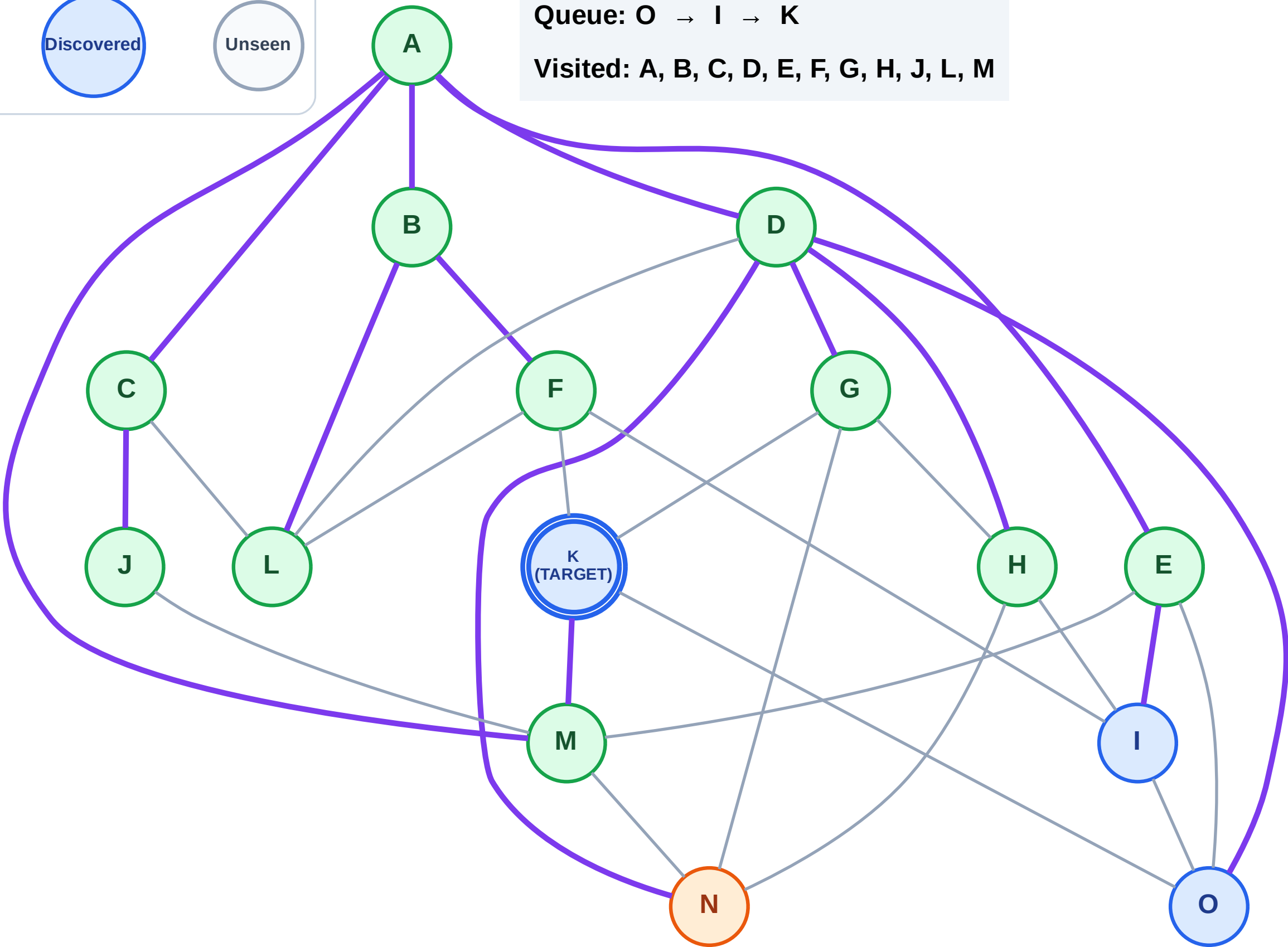
Processing

Discovered

Unseen

Queue: O → I → K

Visited: A, B, C, D, E, F, G, H, J, L, M



BFS Traversal

Step 25: No new neighbors from N

Legend

Complete

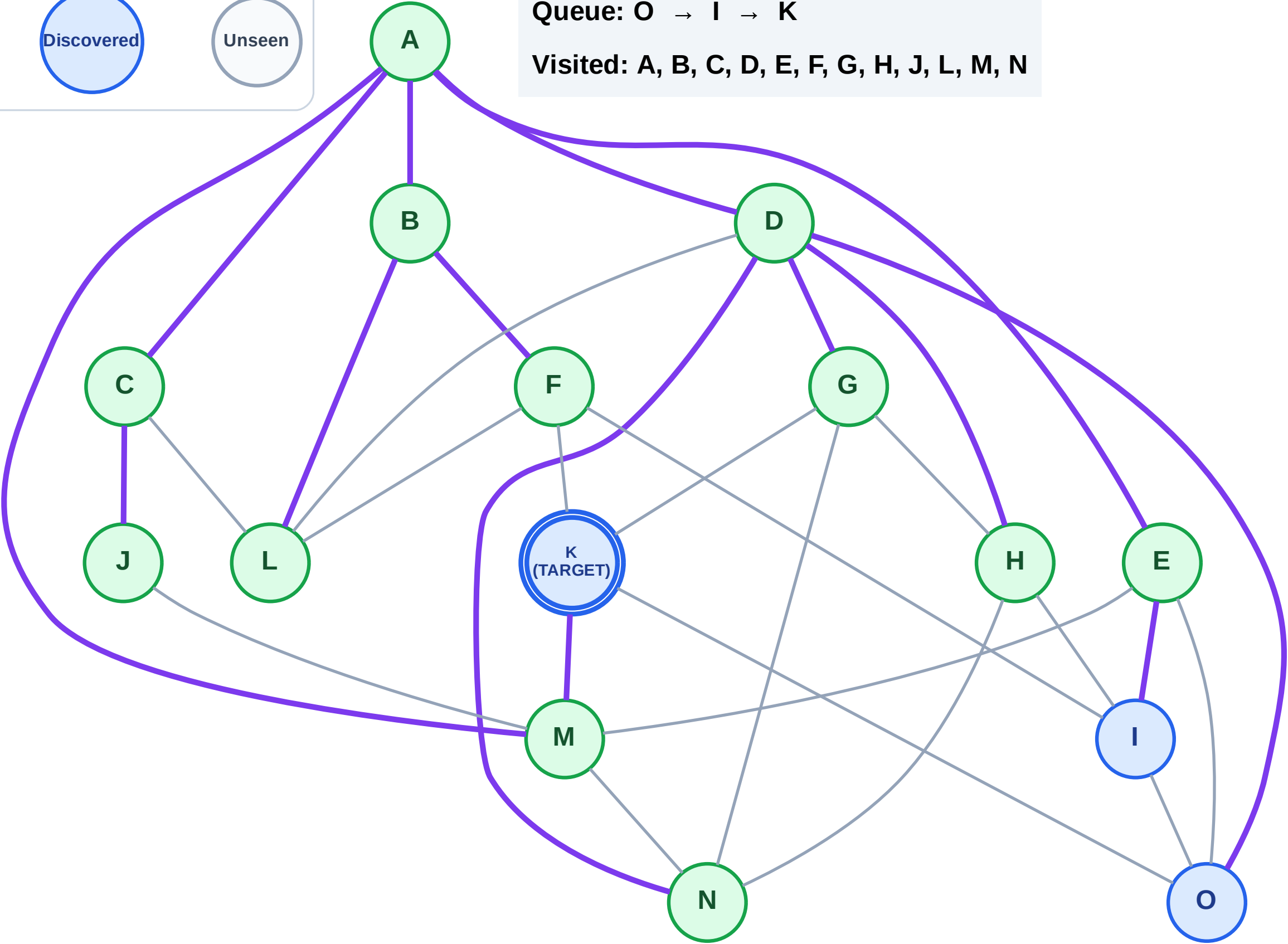
Processing

Discovered

Unseen

Queue: O → I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N



BFS Traversal

Step 26: Process node O

Legend

Complete

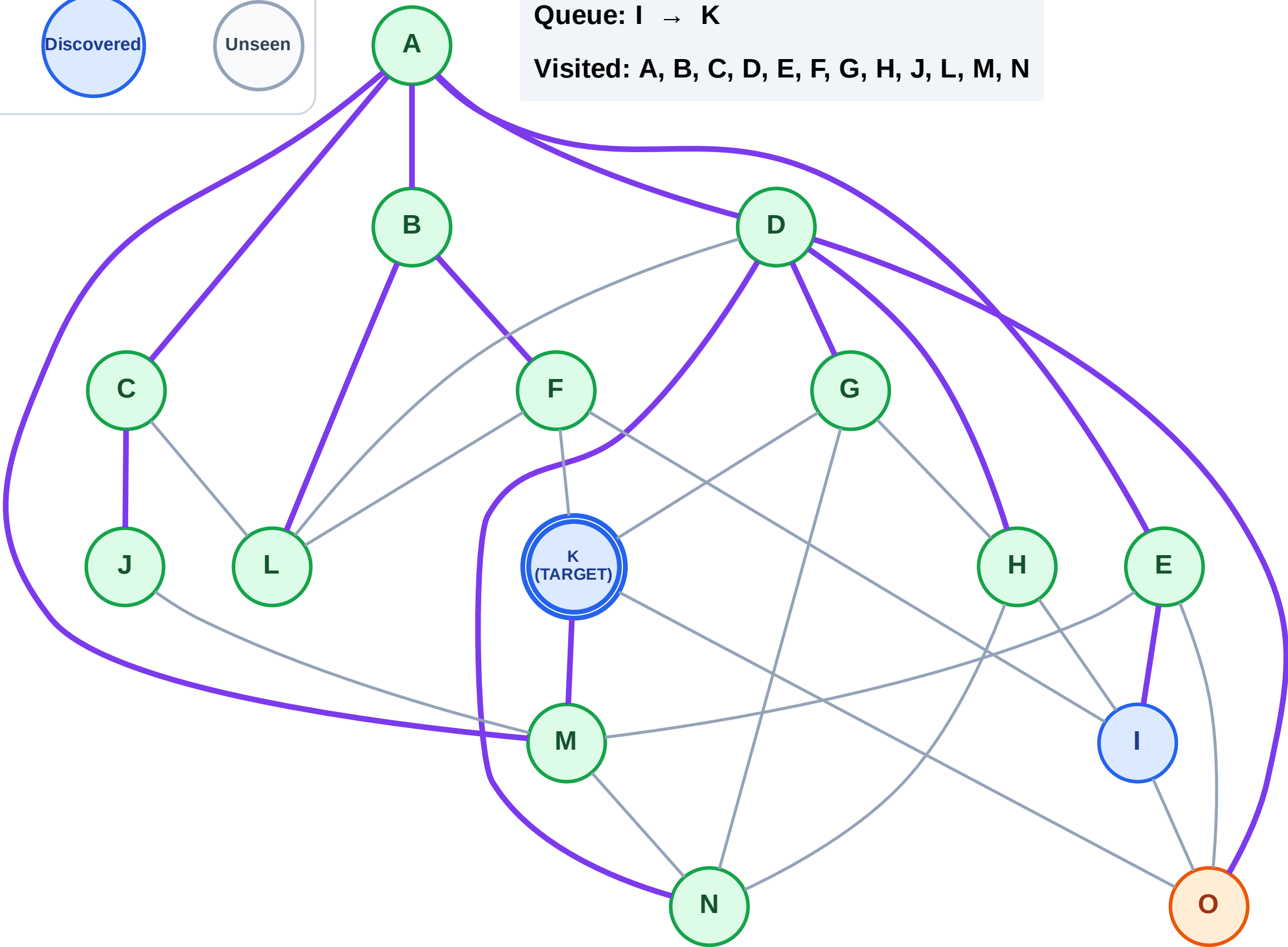
Processing

Discovered

Unseen

Queue: I → K

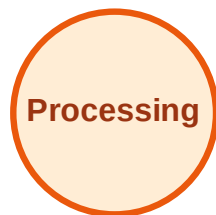
Visited: A, B, C, D, E, F, G, H, J, L, M, N



BFS Traversal

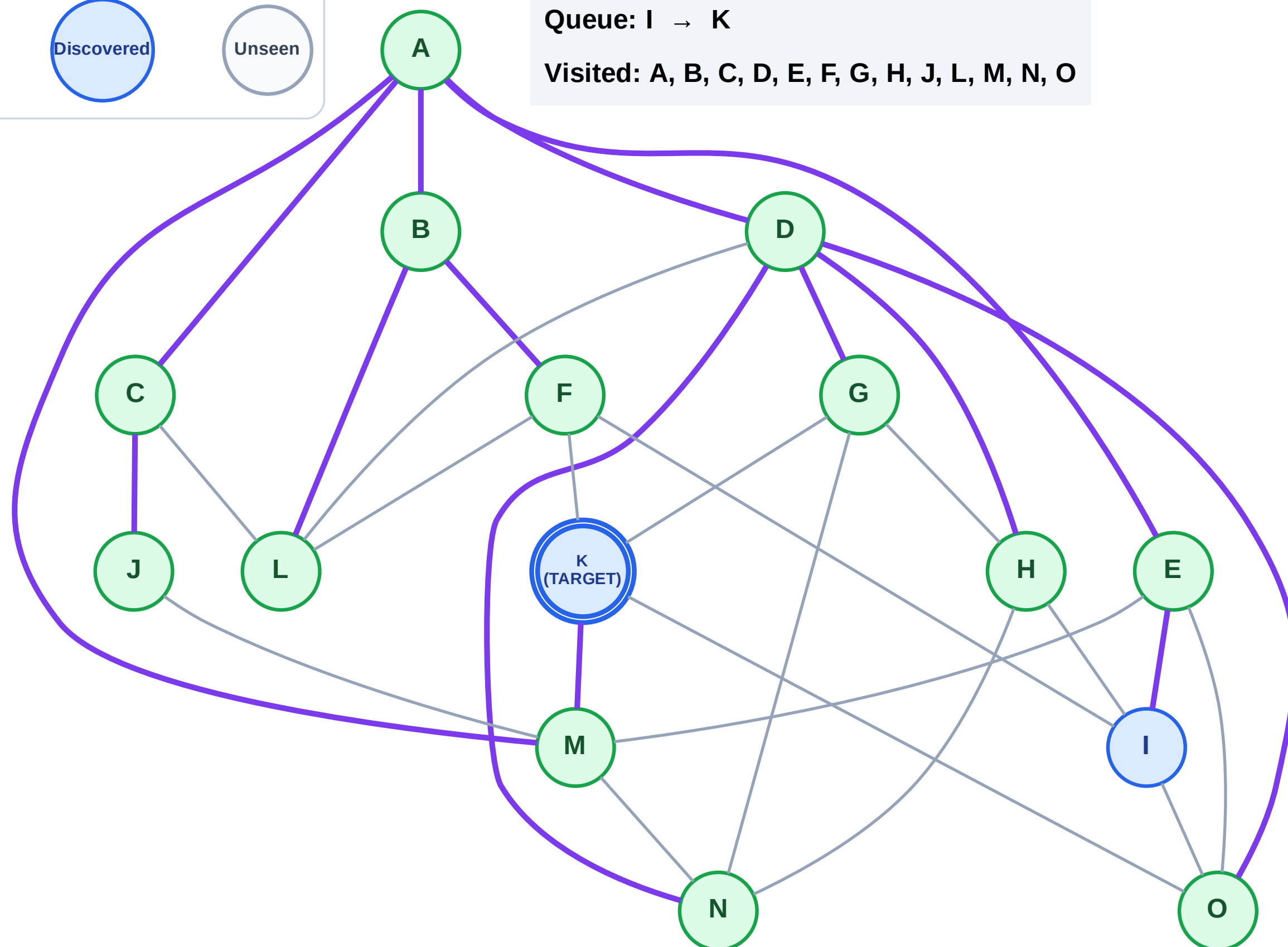
Step 27: No new neighbors from O

Legend



Queue: I → K

Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

Step 28: Process node I

Legend

Complete

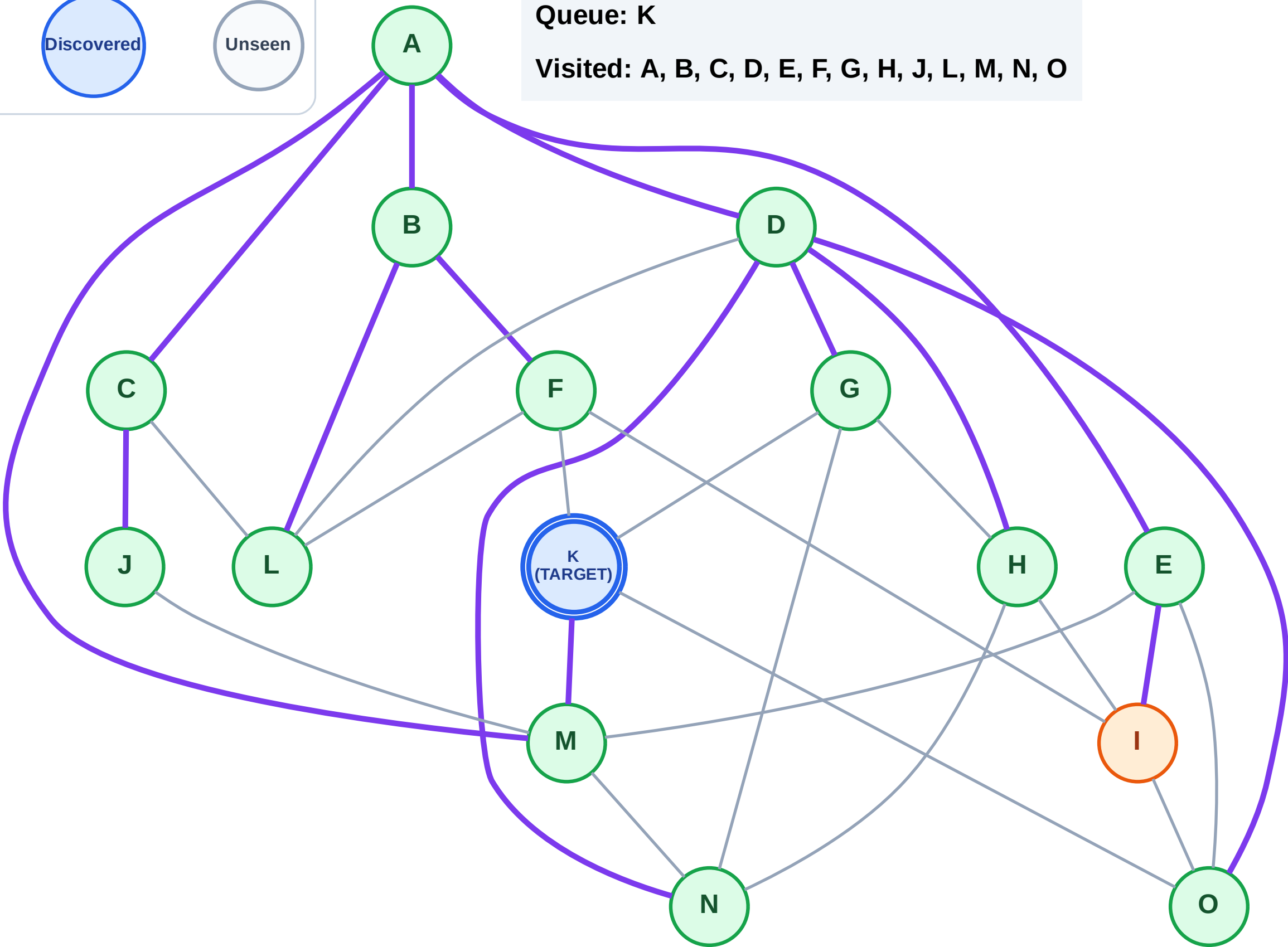
Processing

Discovered

Unseen

Queue: K

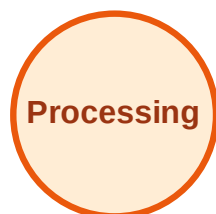
Visited: A, B, C, D, E, F, G, H, J, L, M, N, O



BFS Traversal

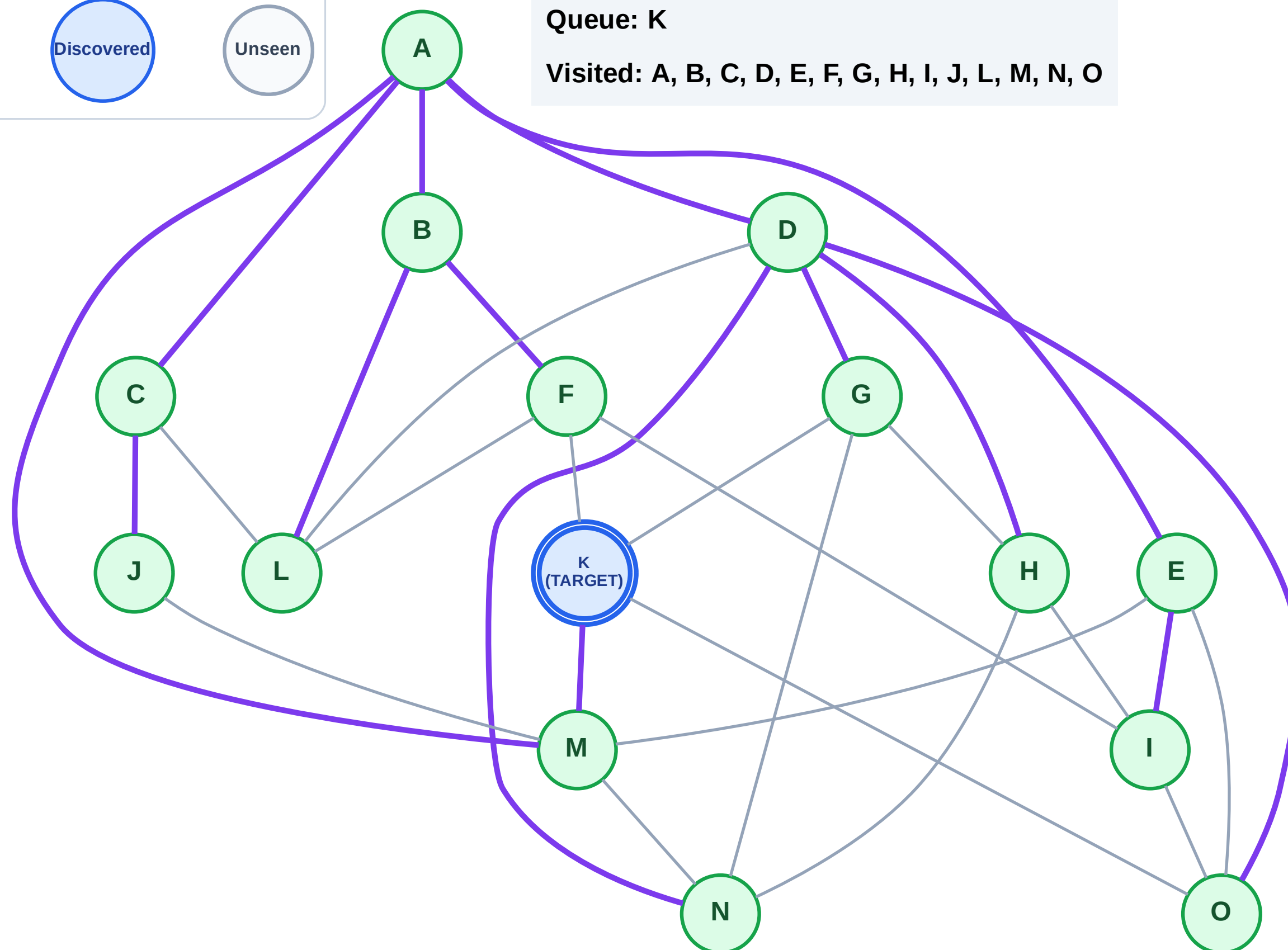
Step 29: No new neighbors from I

Legend



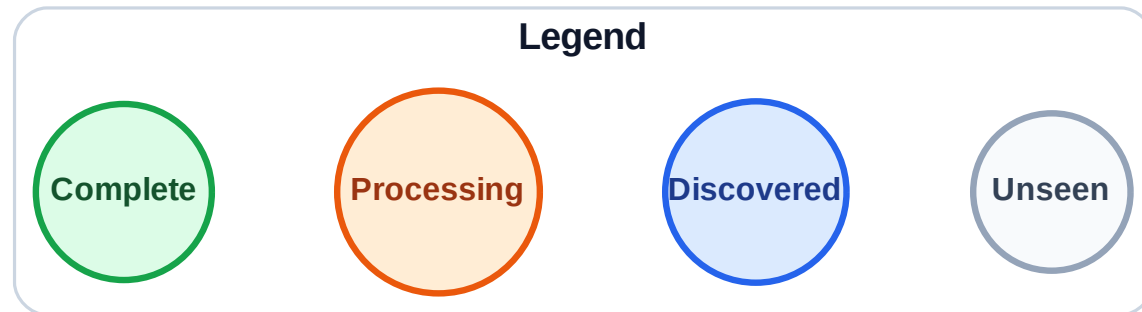
Queue: K

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



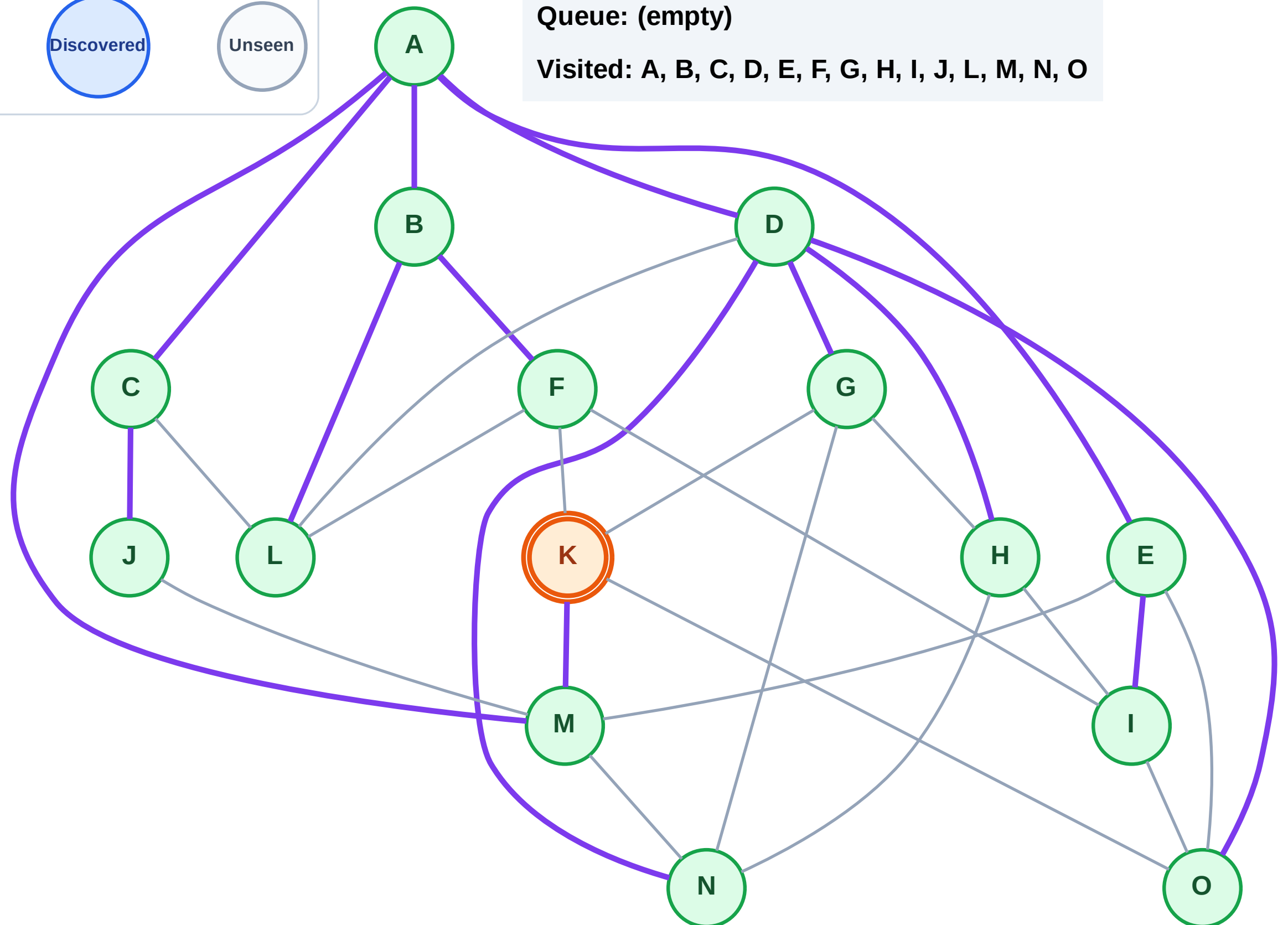
Step 30: Process node K

Step 30: Process node K



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N, O



BFS Traversal

Step 31: TARGET NODE K REACHED! Path (2 hops): A → M → K

Legend

Complete

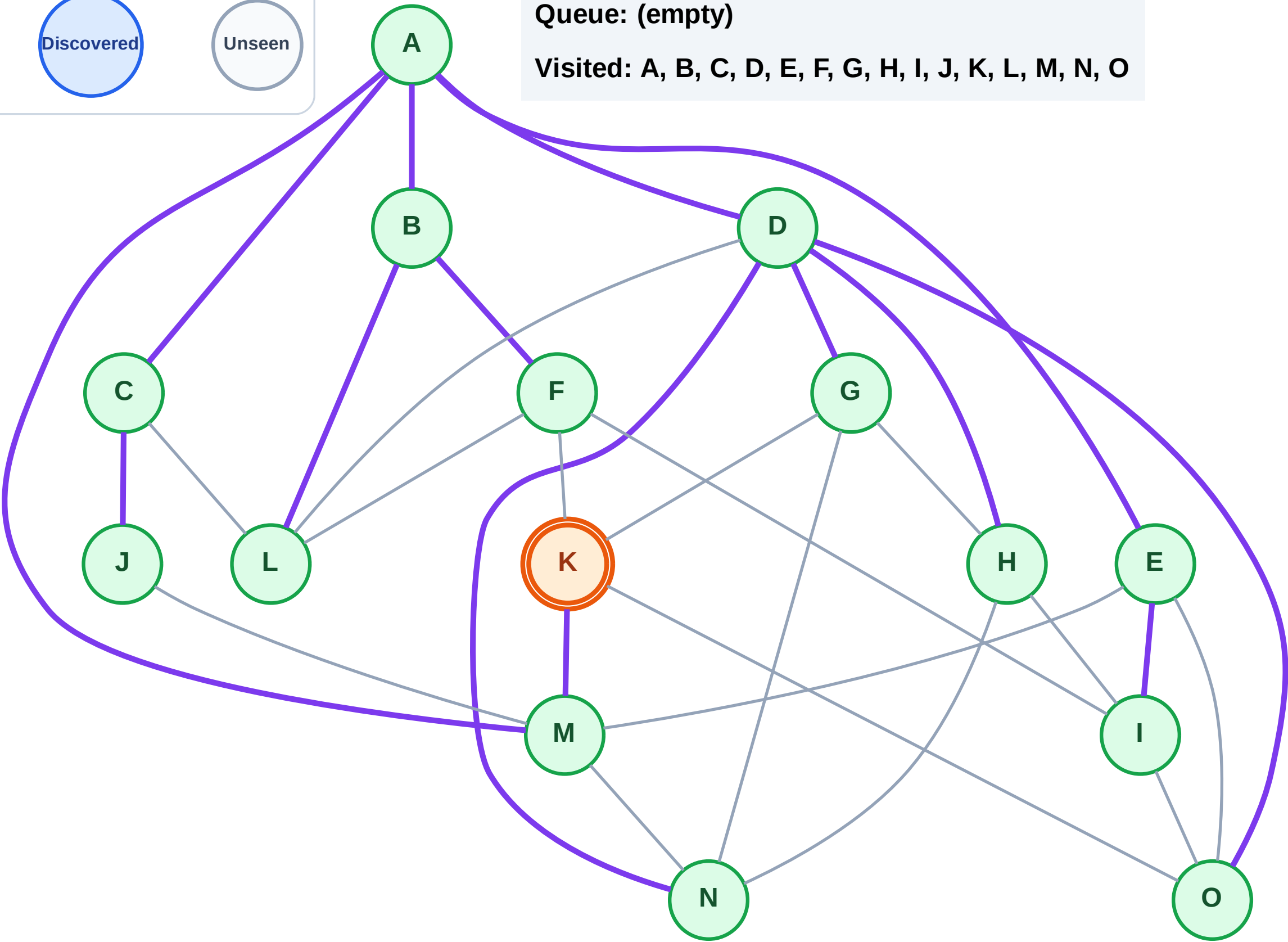
Processing

Discovered

Unseen

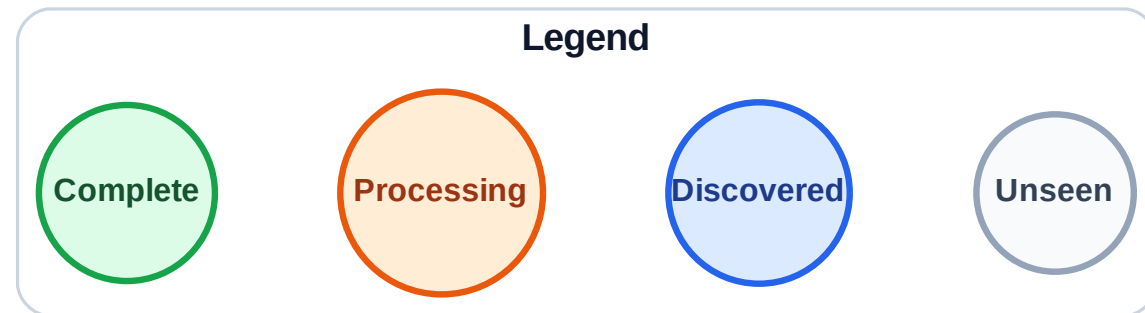
Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N, O

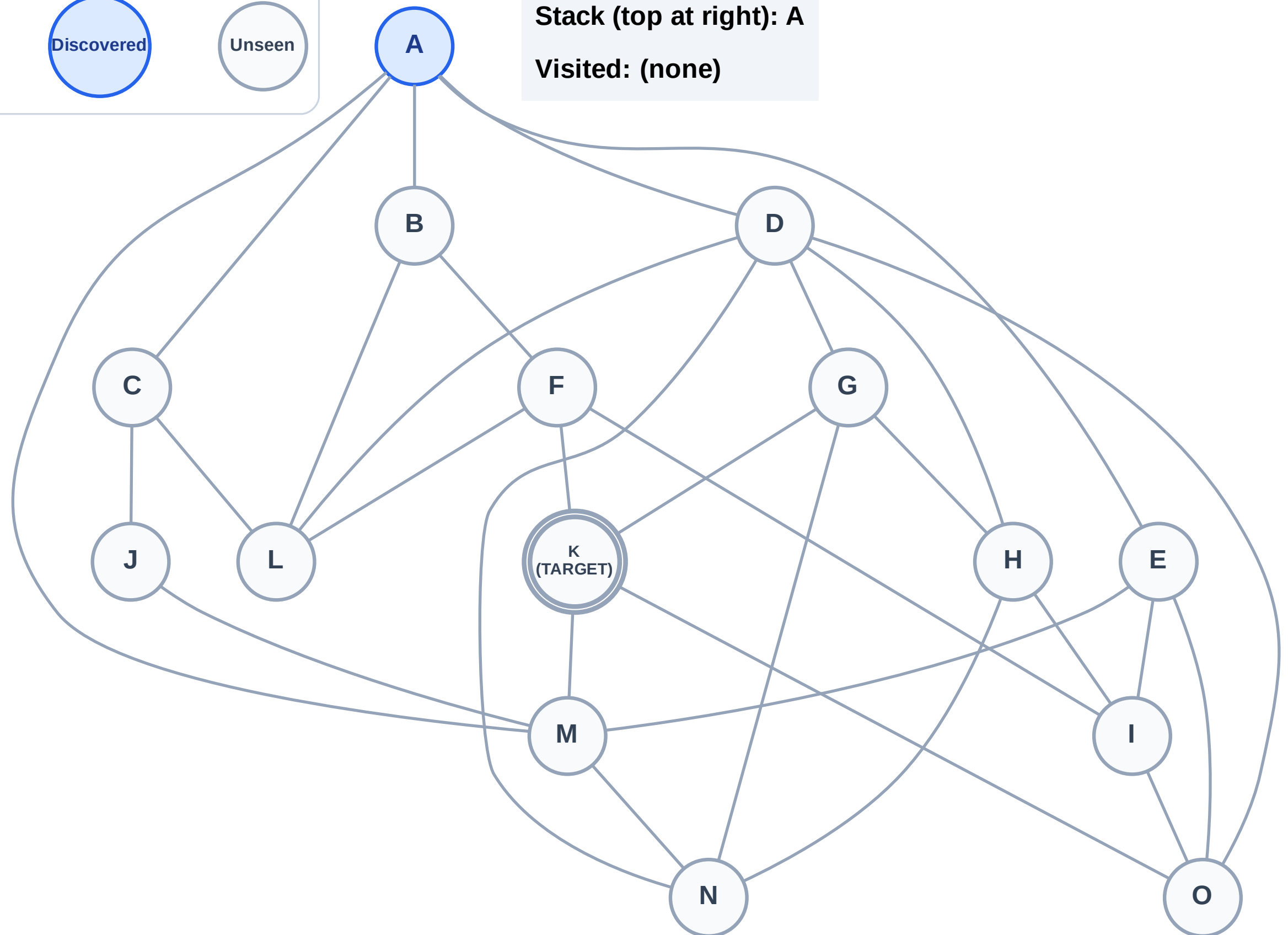


DFS Traversal

Step 1: Start at A (Seeking target node K)



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

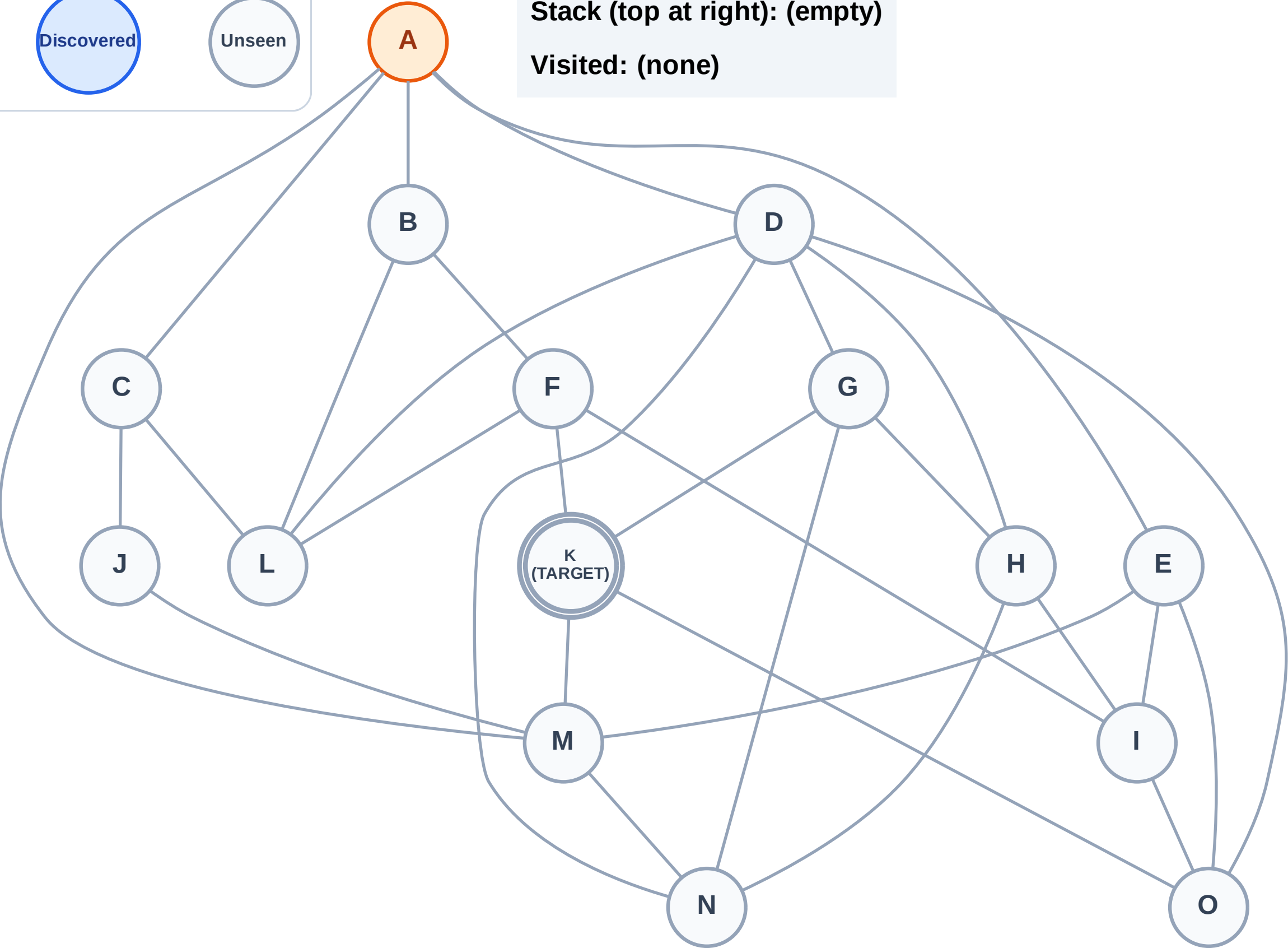
Complete

Processing

Discovered

Unseen

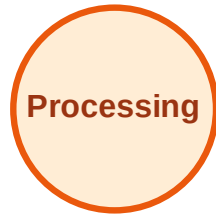
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

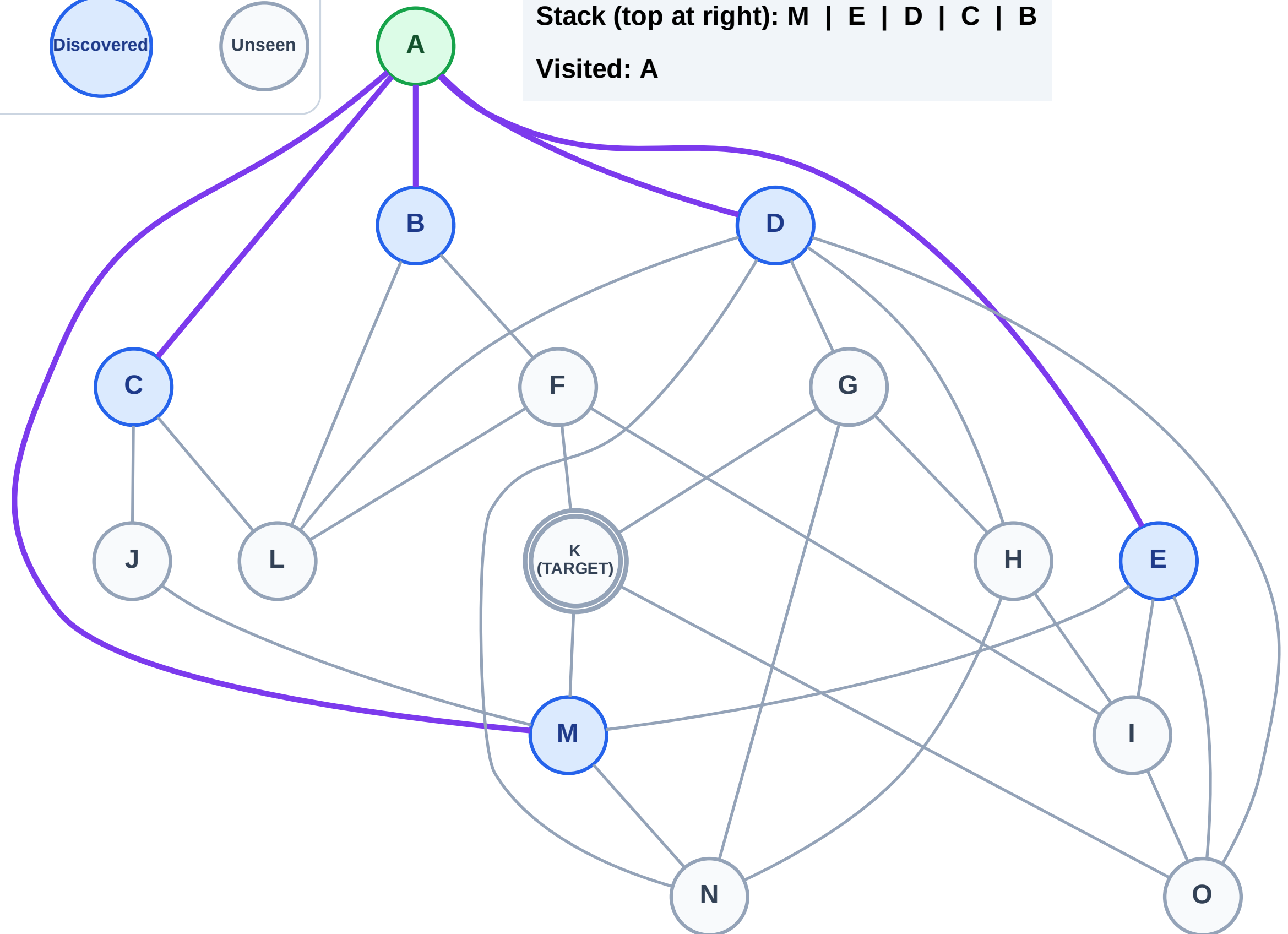
Step 3: Discover M, E, D, C, B from A

Legend



Stack (top at right): M | E | D | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

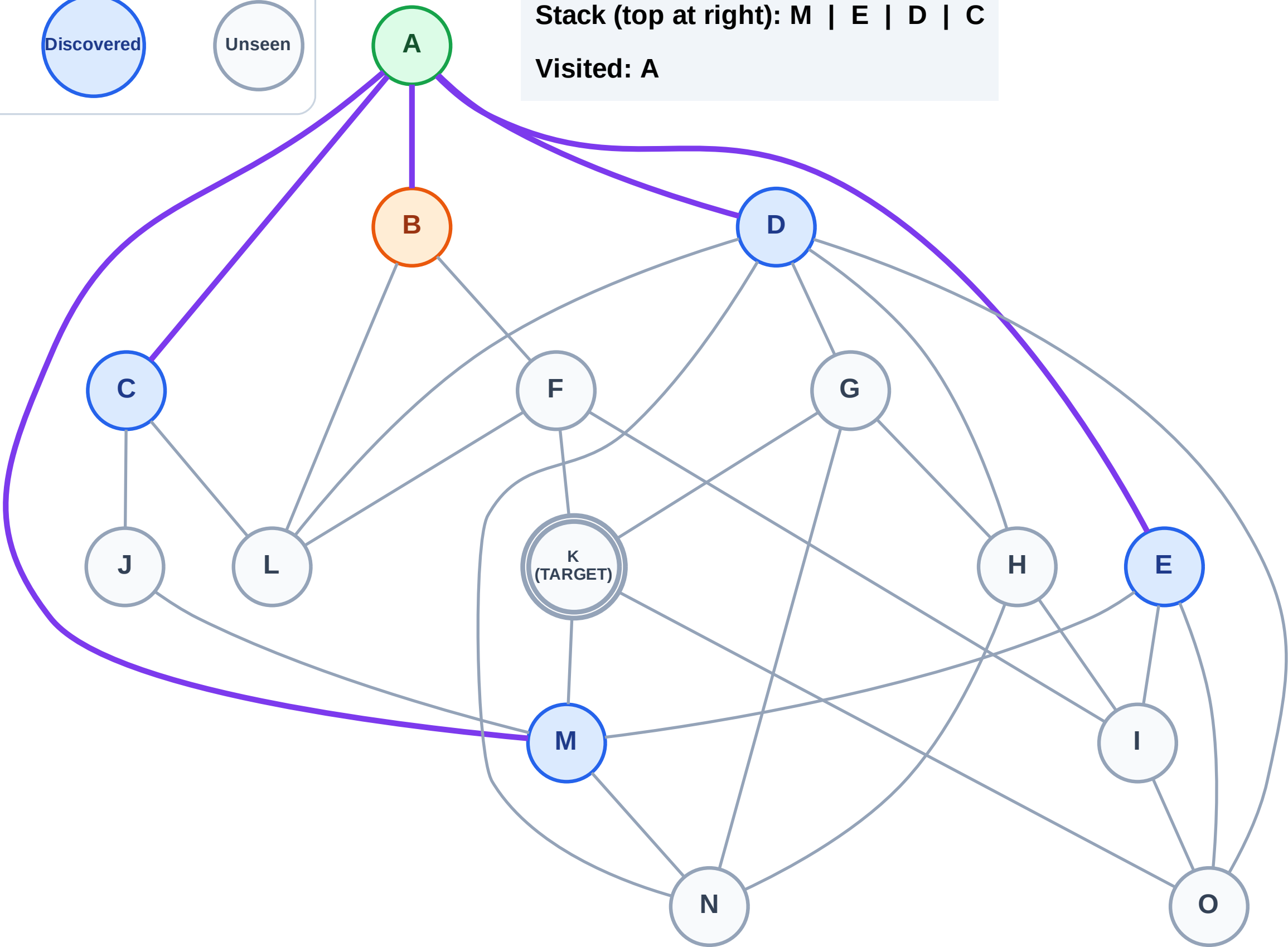
Complete

Processing

Discovered

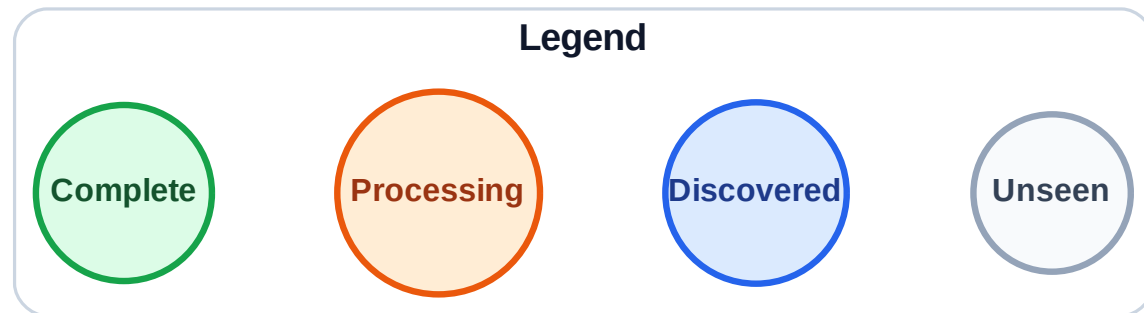
Unseen

Stack (top at right): M | E | D | C
Visited: A



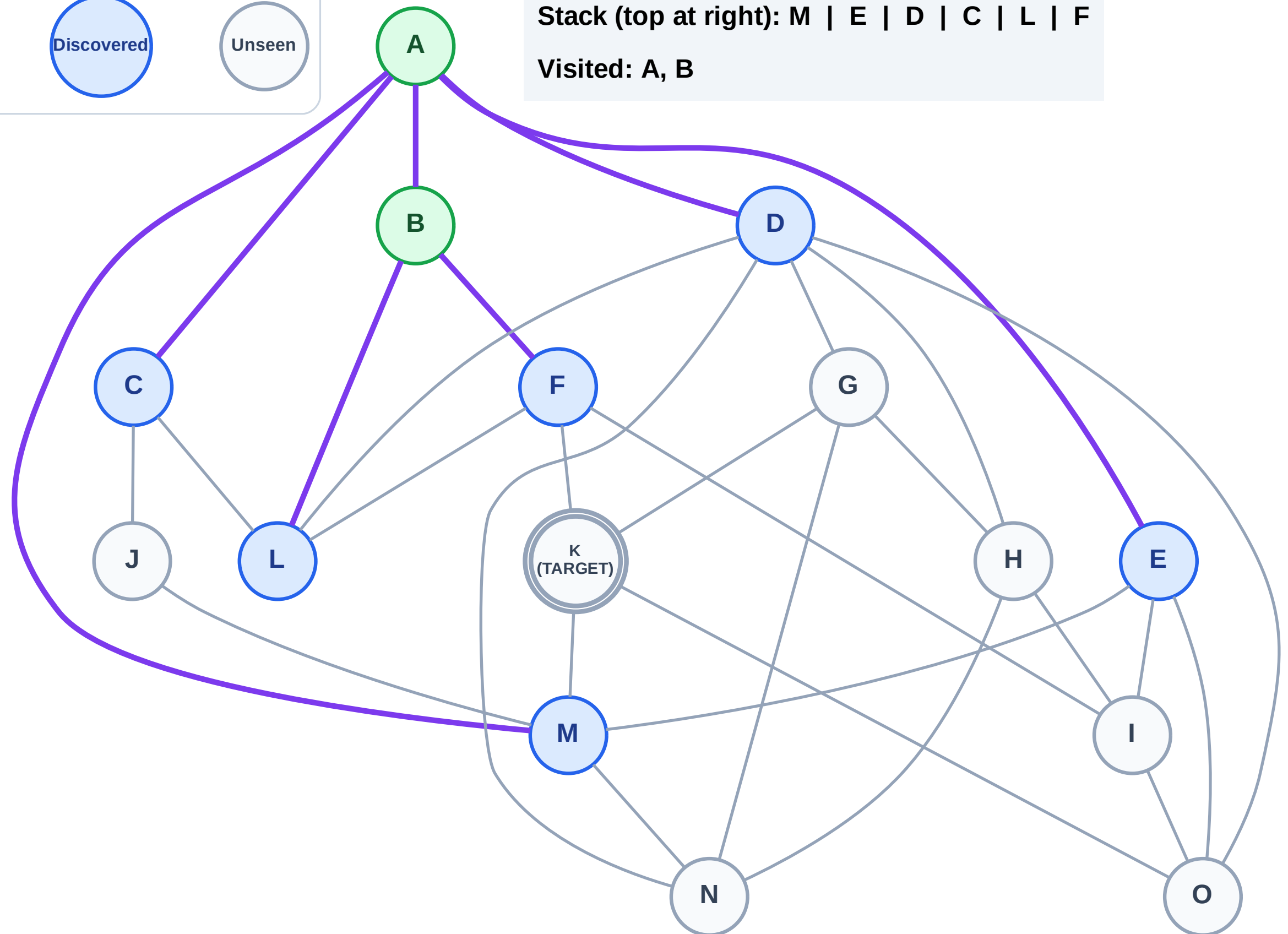
Step 5: Discover L, F from B

Step 5: Discover L, F from B



Stack (top at right): M | E | D | C | L | F

Visited: A, B



DFS Traversal

Step 6: Process node F

Legend

Complete

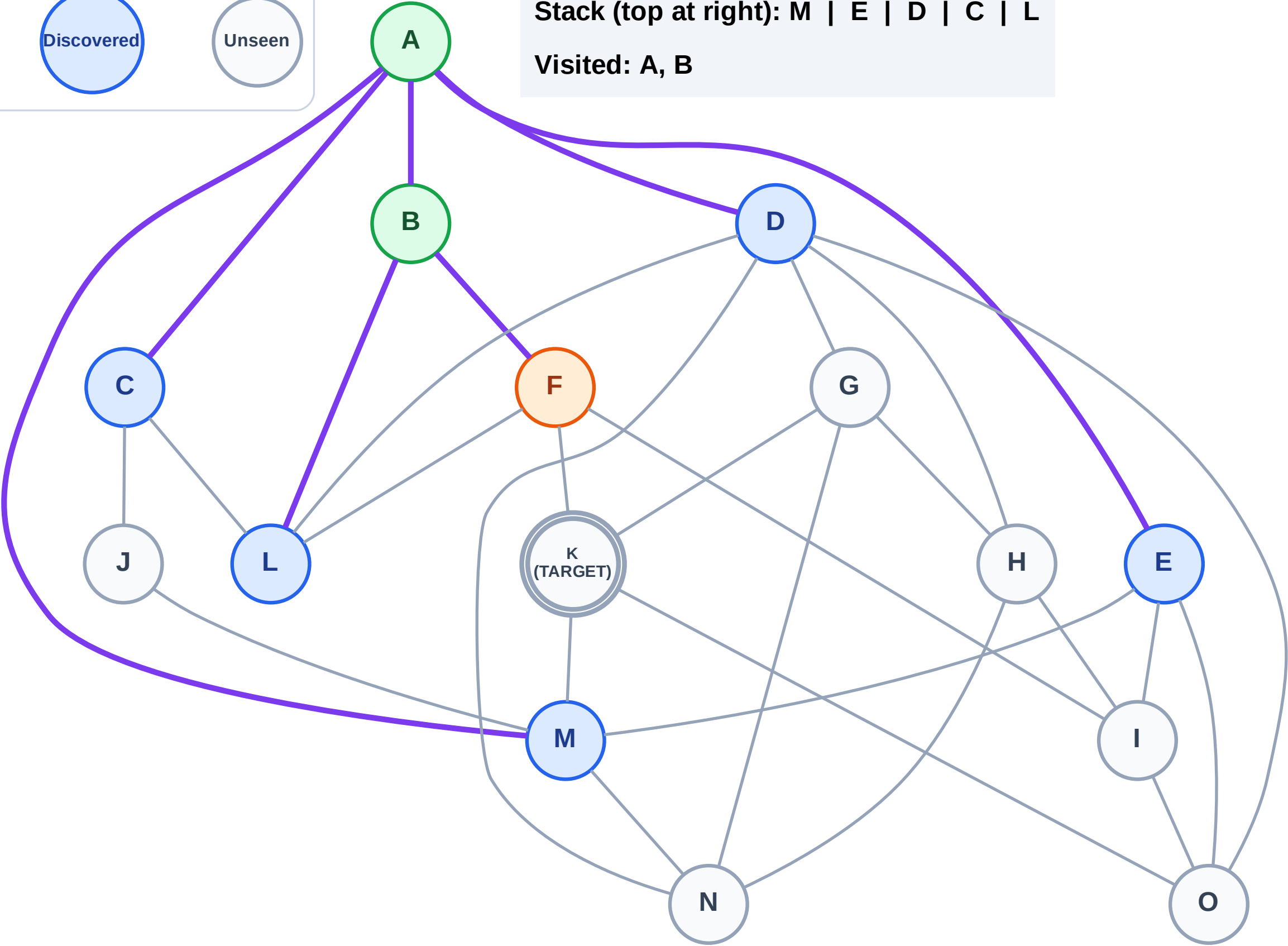
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L

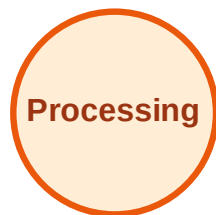
Visited: A, B



DFS Traversal

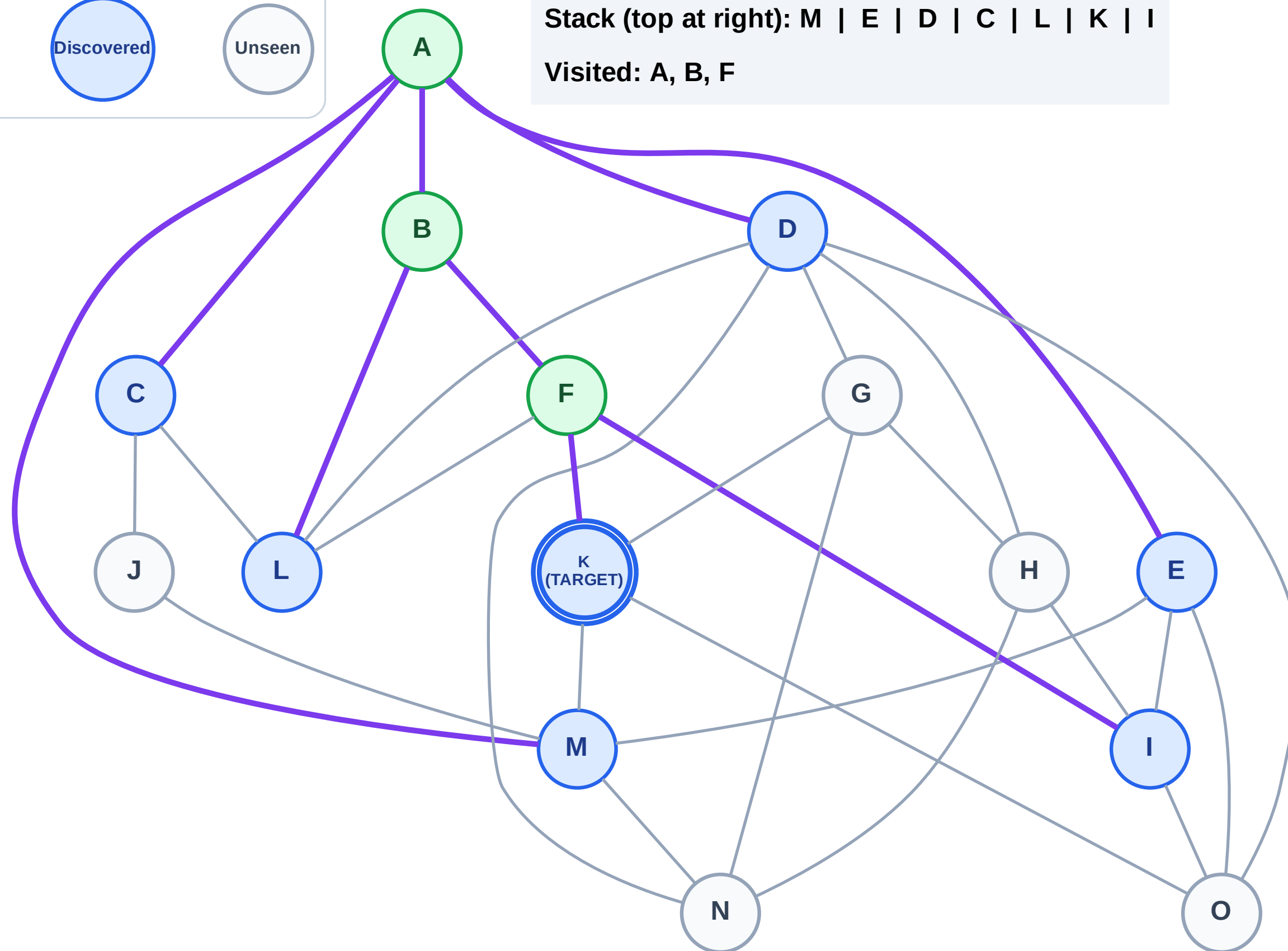
Step 7: Discover K, I from F

Legend



Stack (top at right): M | E | D | C | L | K | I

Visited: A, B, F



DFS Traversal

Step 8: Process node I

Legend

Complete

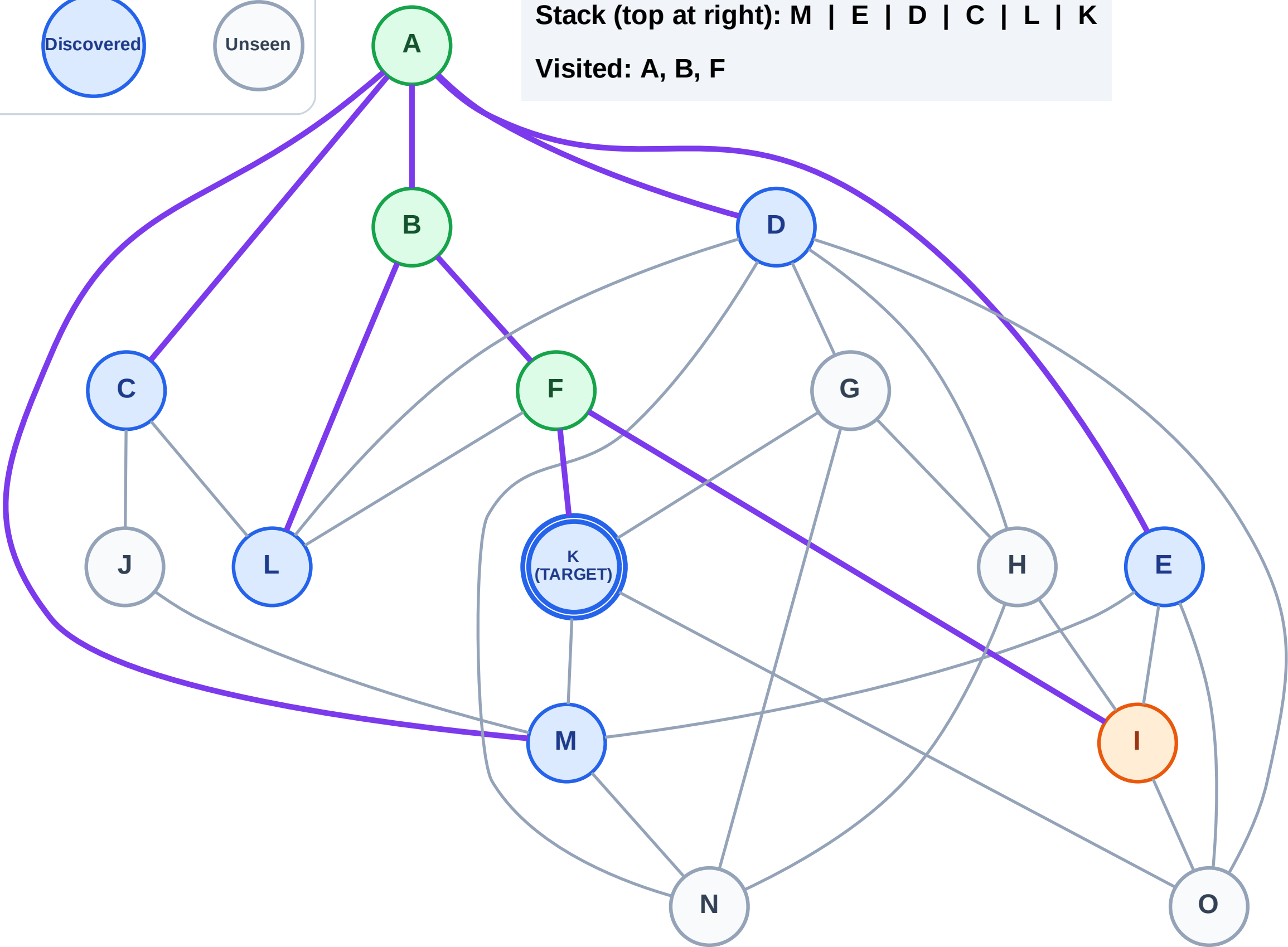
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K

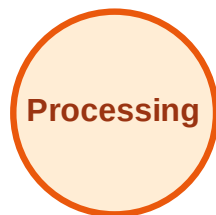
Visited: A, B, F



DFS Traversal

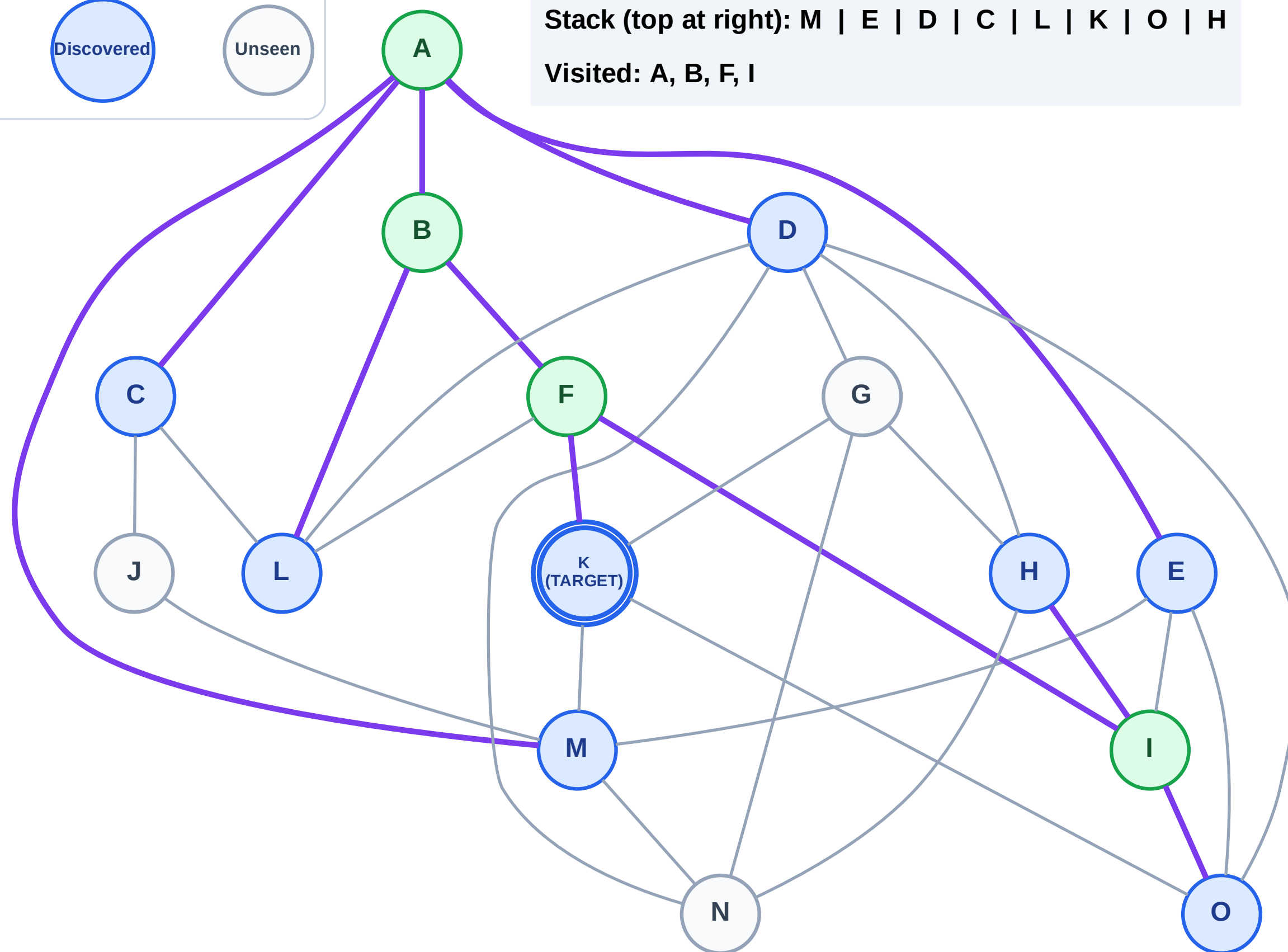
Step 9: Discover O, H from I

Legend



Stack (top at right): M | E | D | C | L | K | O | H

Visited: A, B, F, I



DFS Traversal

Step 10: Process node H

Legend

Complete

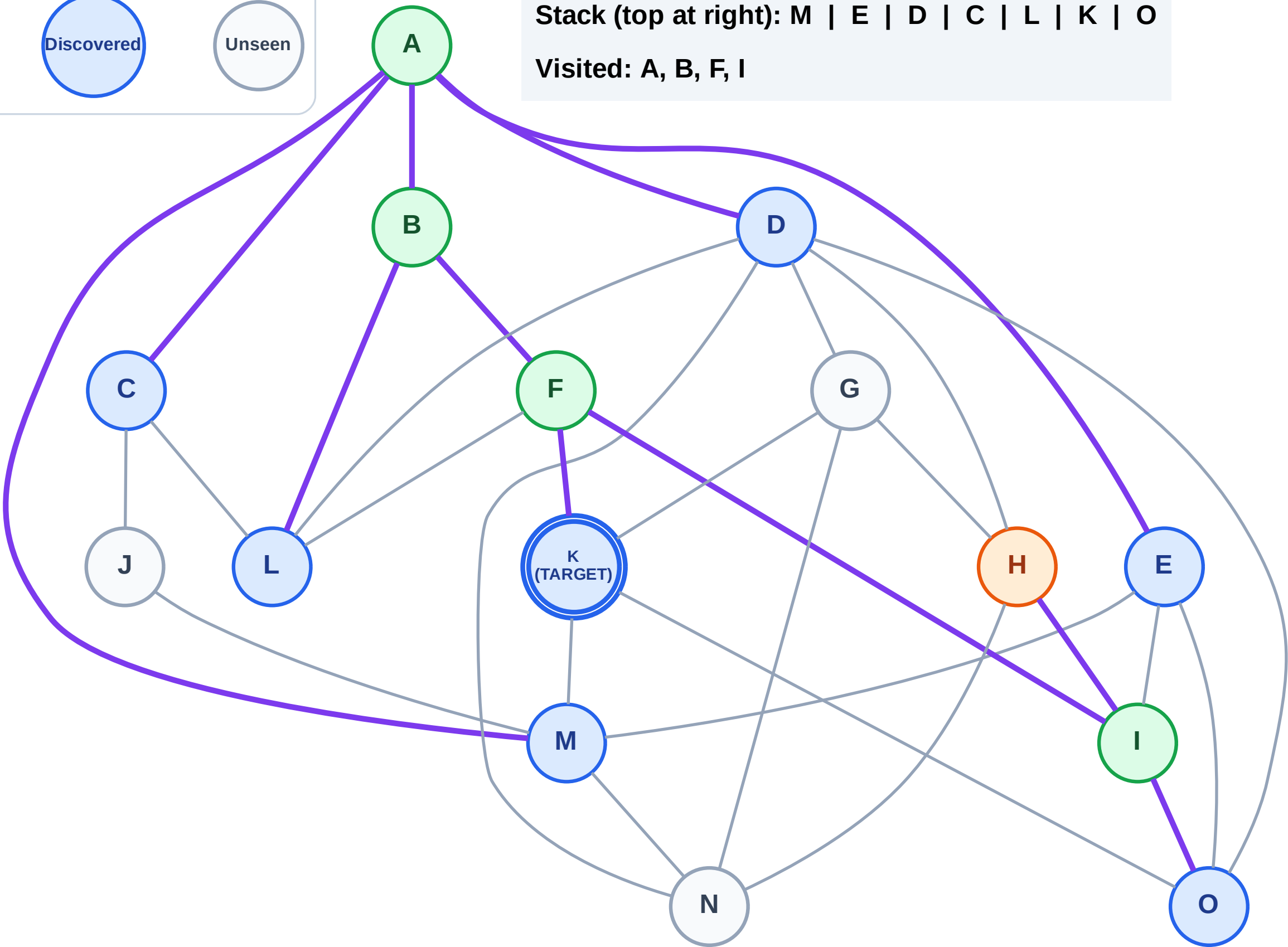
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O

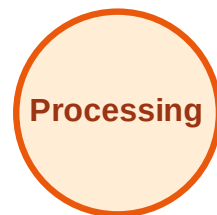
Visited: A, B, F, I



DFS Traversal

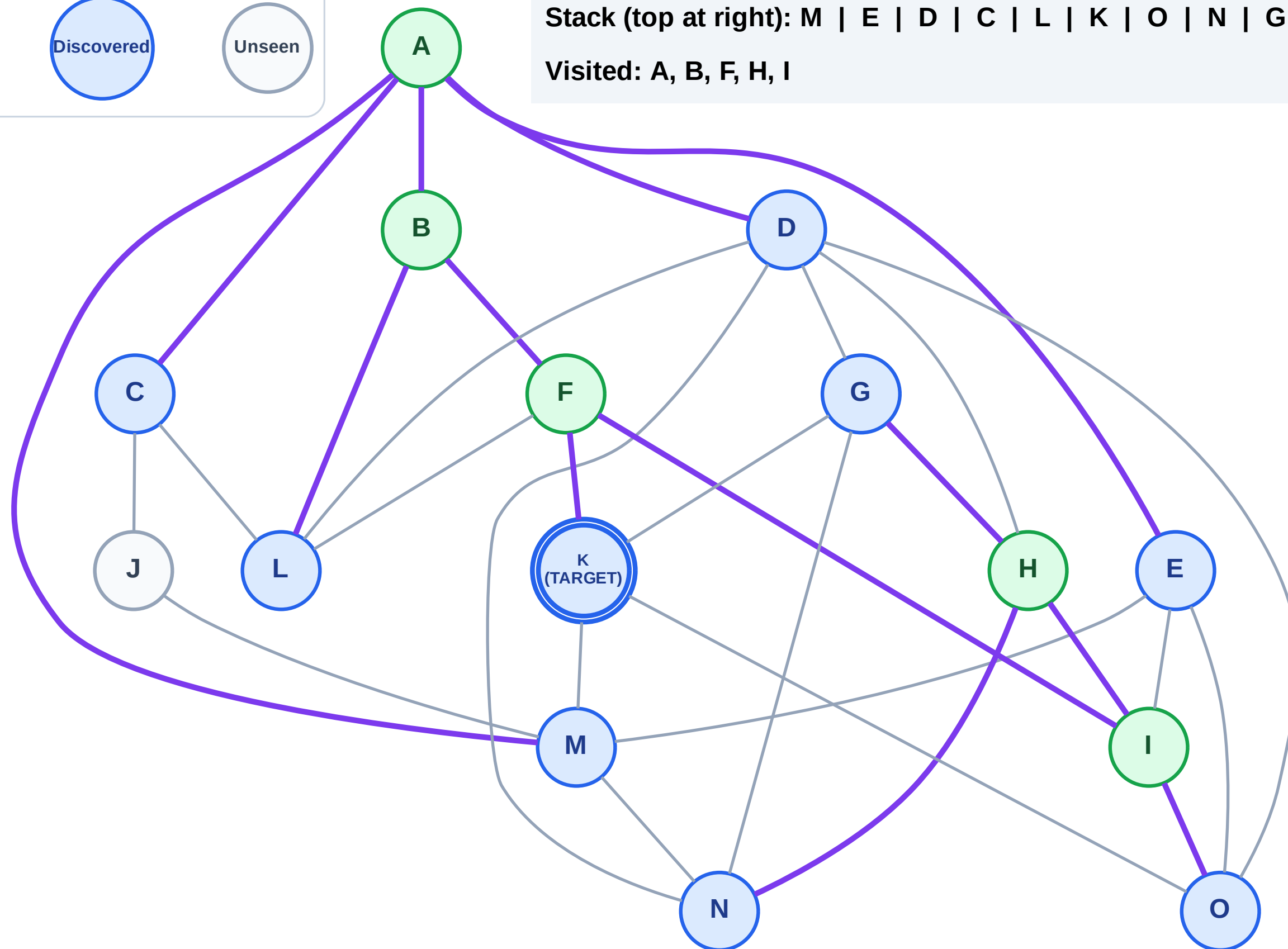
Step 11: Discover N, G from H

Legend



Stack (top at right): M | E | D | C | L | K | O | N | G

Visited: A, B, F, H, I



DFS Traversal

Step 12: Process node G

Legend

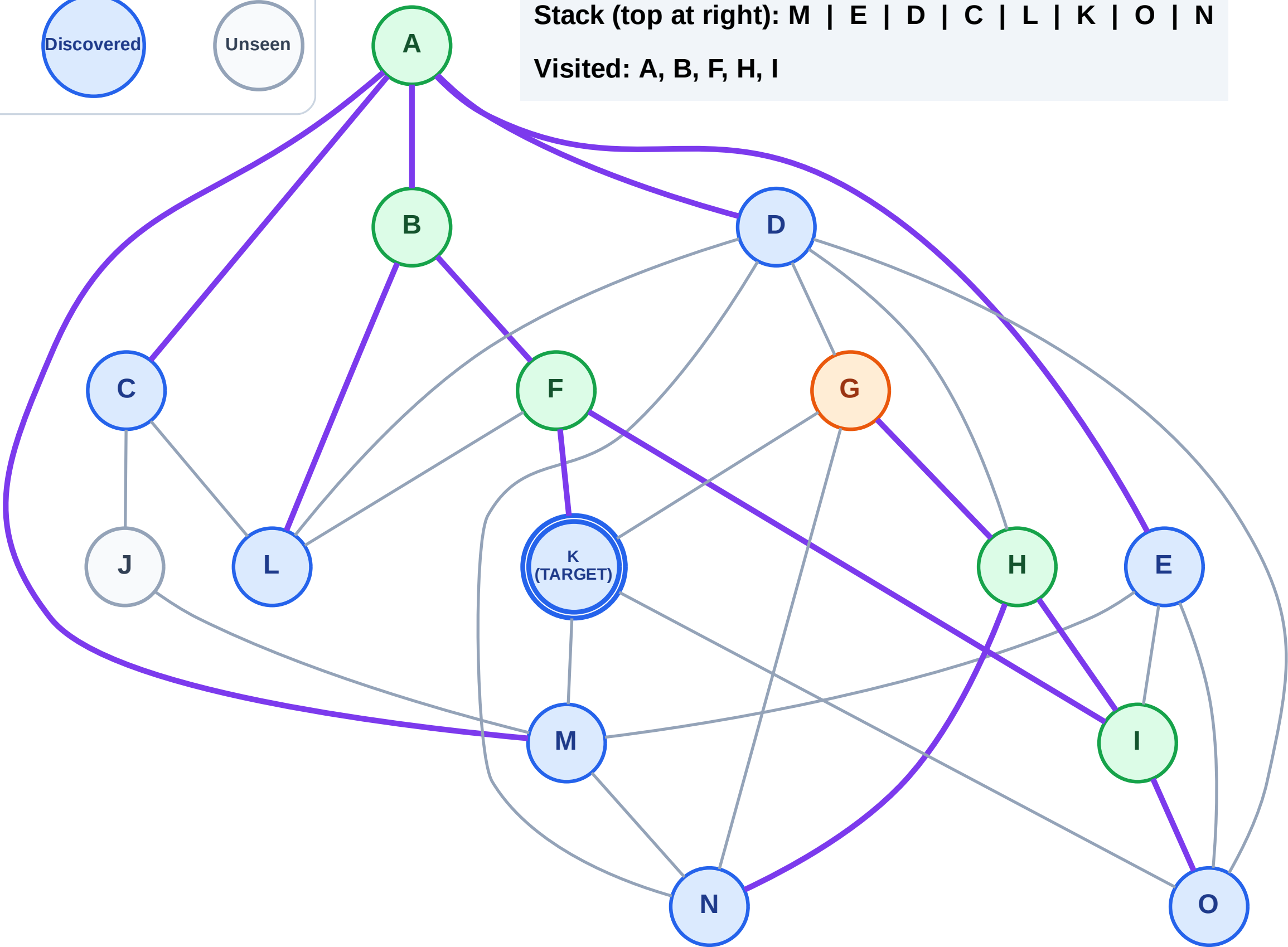
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O | N
Visited: A, B, F, H, I



DFS Traversal

Step 13: No new neighbors from G

Legend

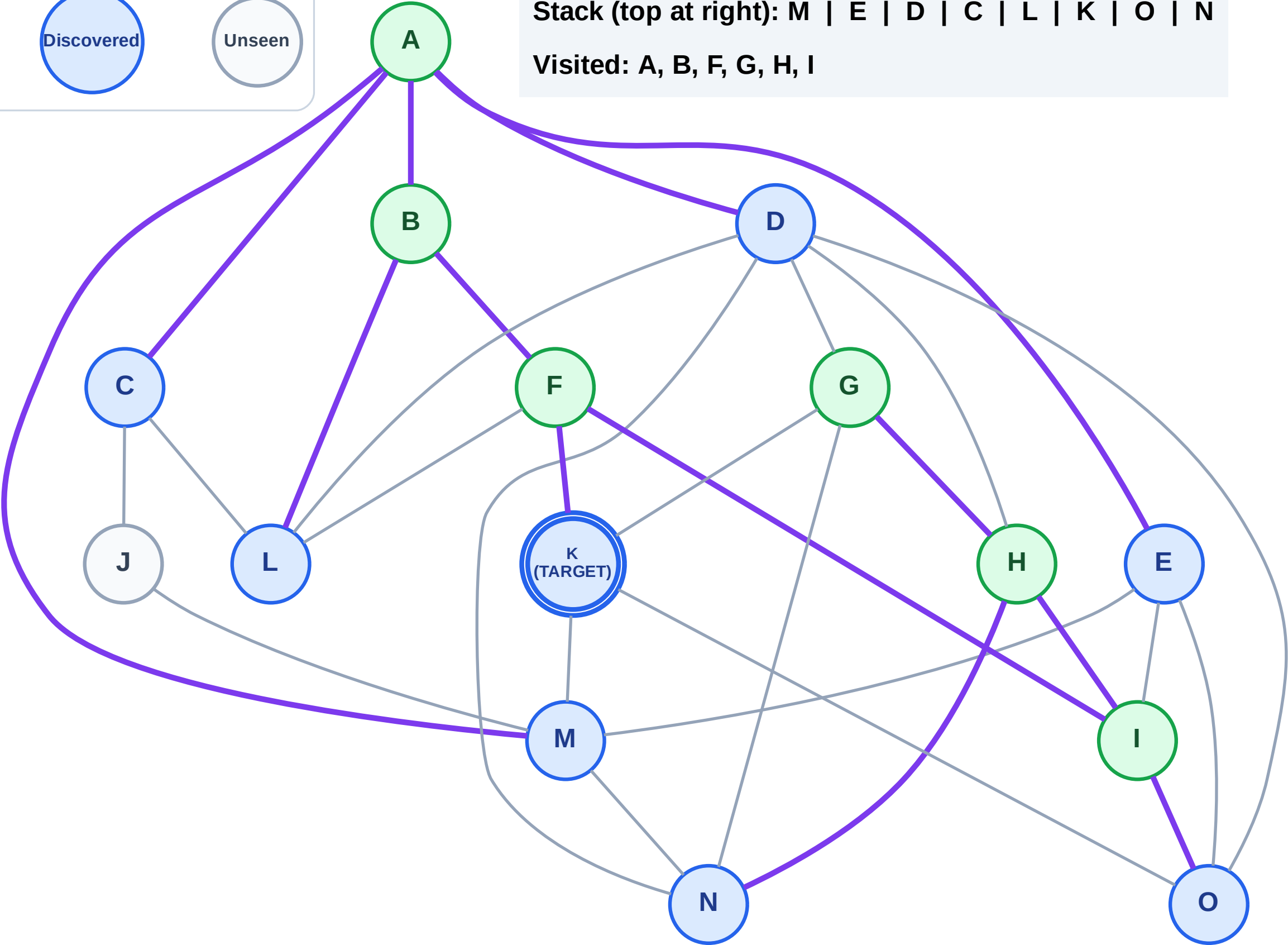
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O | N
Visited: A, B, F, G, H, I



DFS Traversal

Step 14: Process node N

Legend

Complete

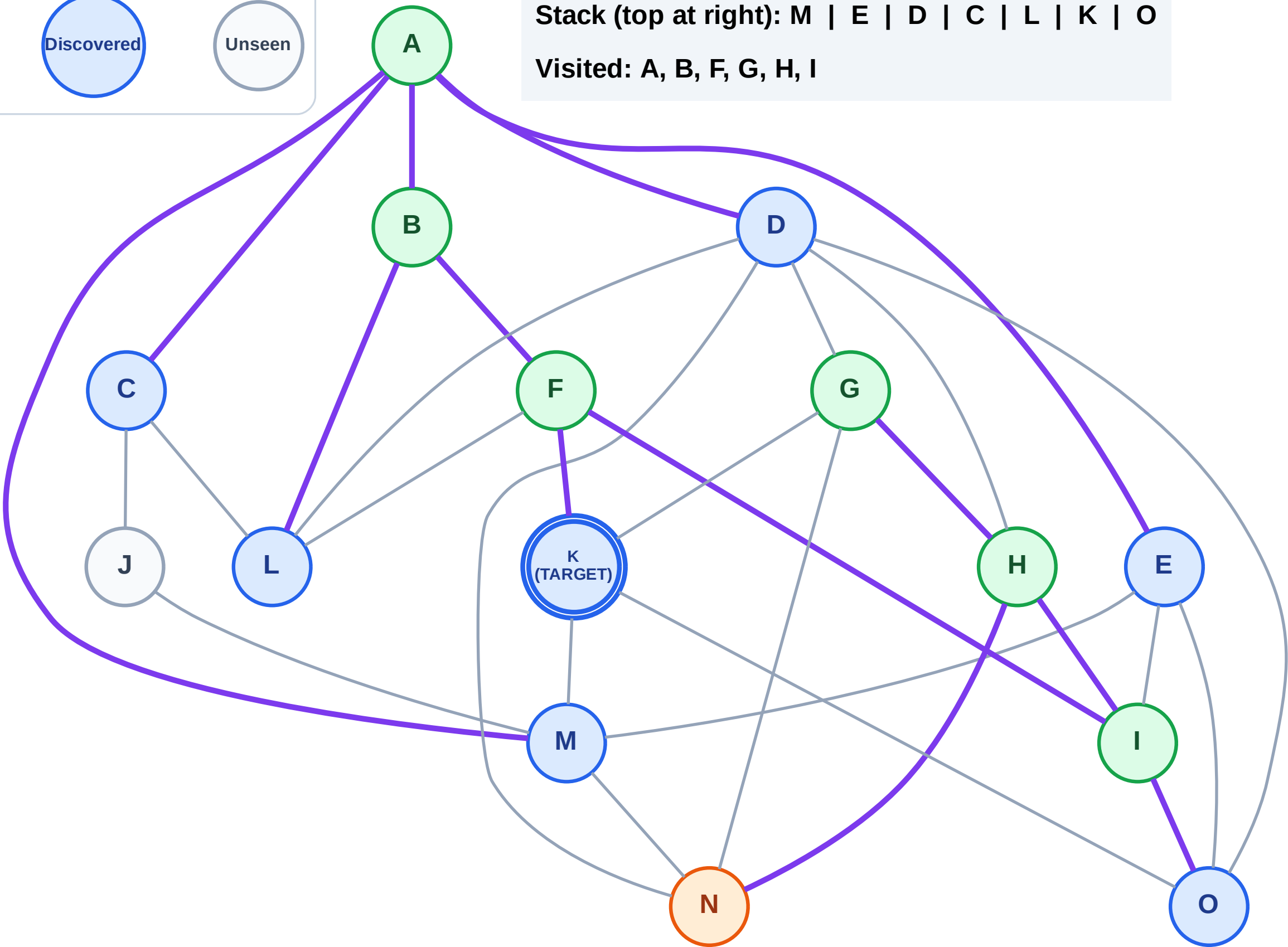
Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K | O

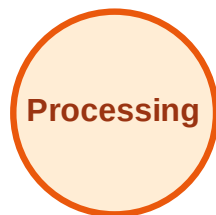
Visited: A, B, F, G, H, I



DFS Traversal

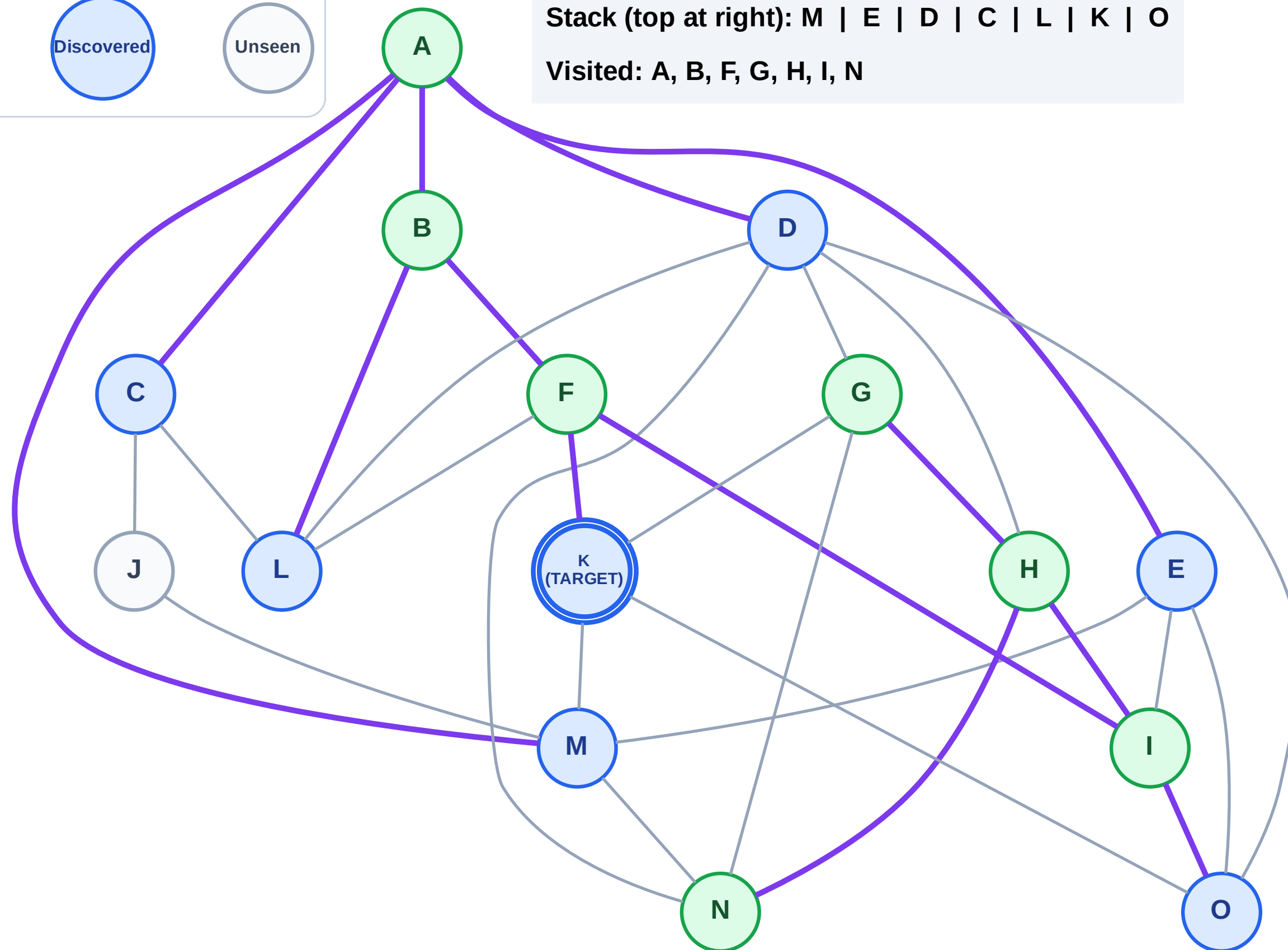
Step 15: No new neighbors from N

Legend



Stack (top at right): M | E | D | C | L | K | O

Visited: A, B, F, G, H, I, N



DFS Traversal

Step 16: Process node O

Legend

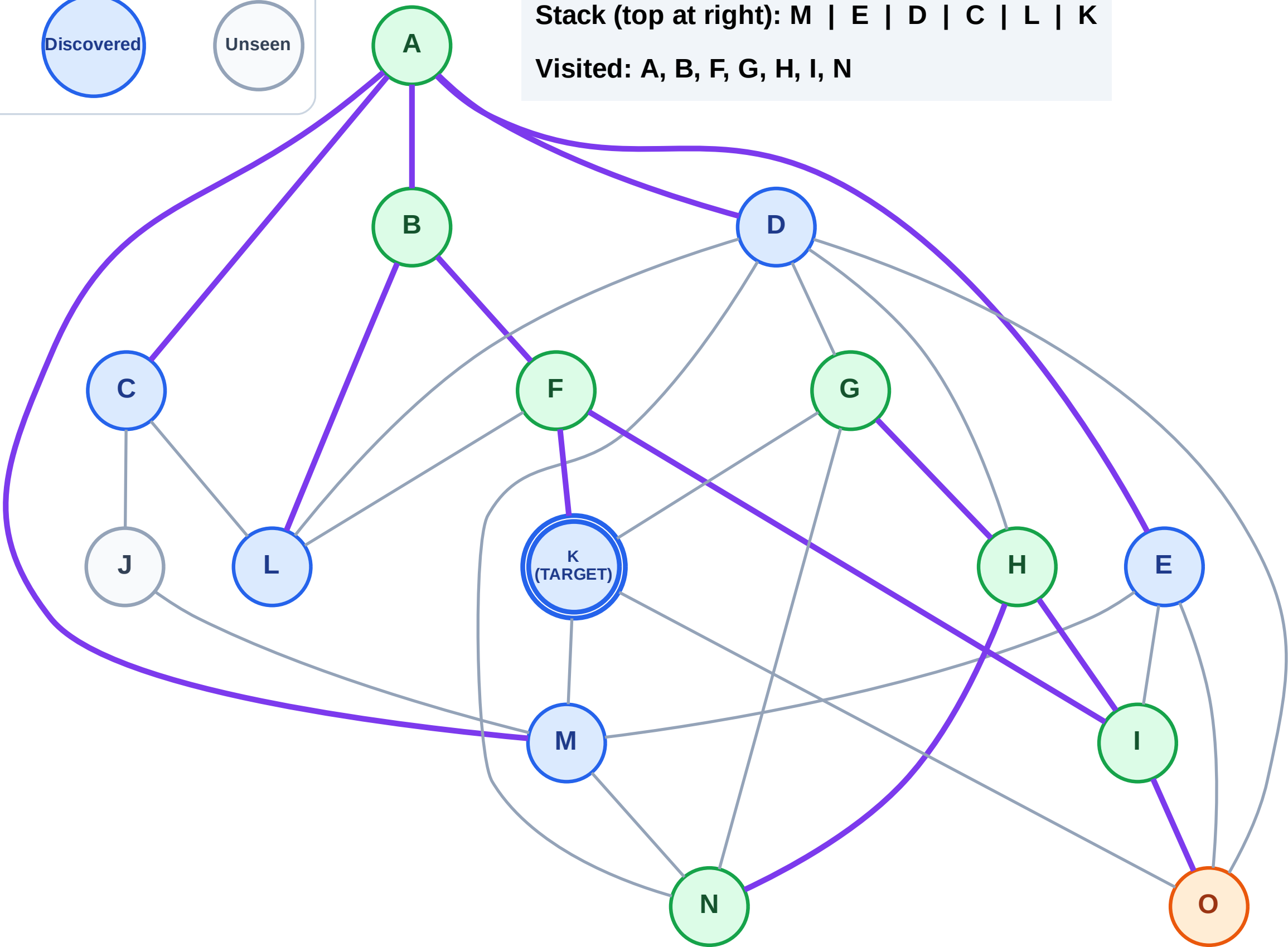
Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L | K
Visited: A, B, F, G, H, I, N



DFS Traversal

Step 17: No new neighbors from O

Legend

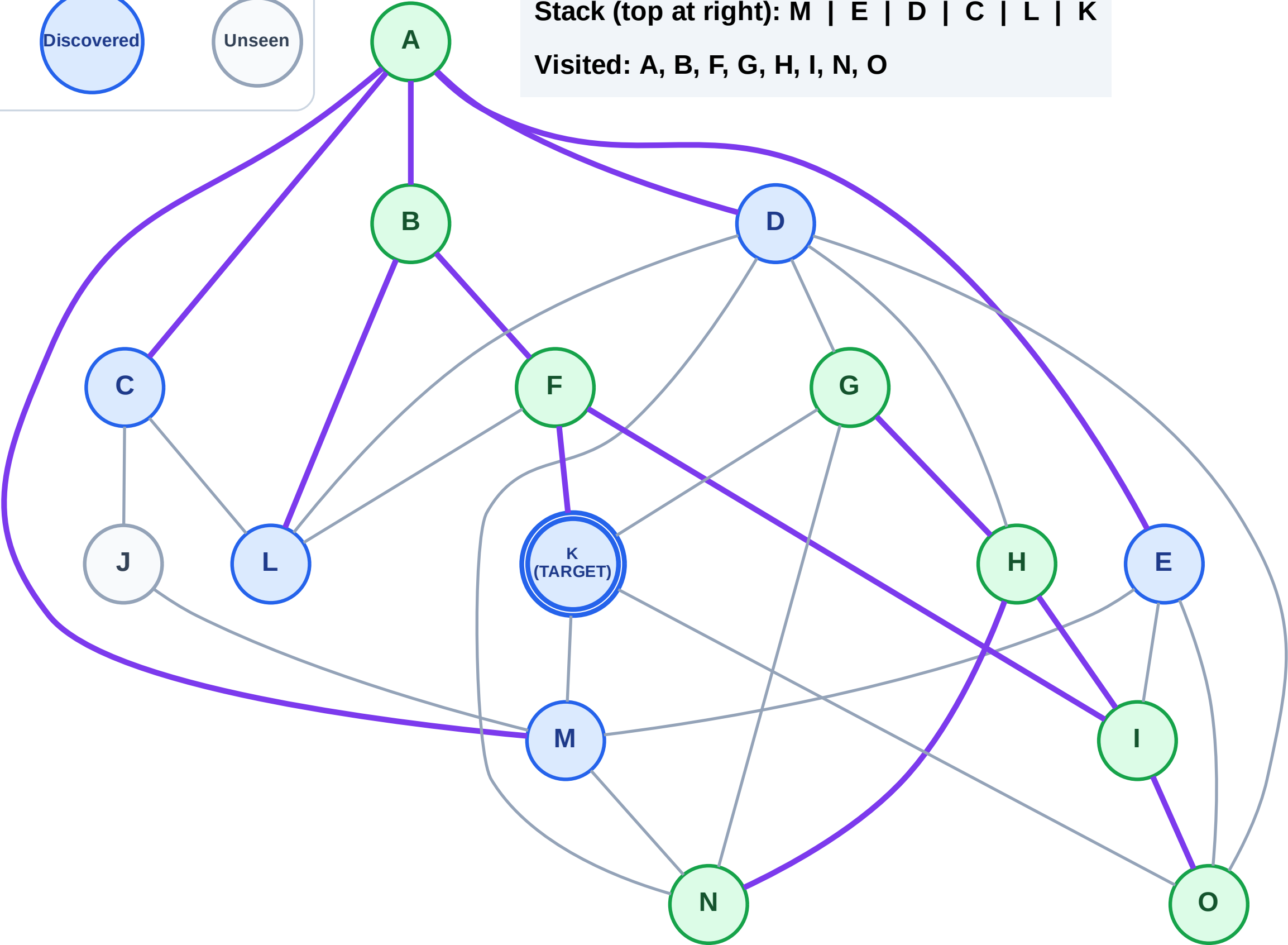
Complete

Processing

Discovered

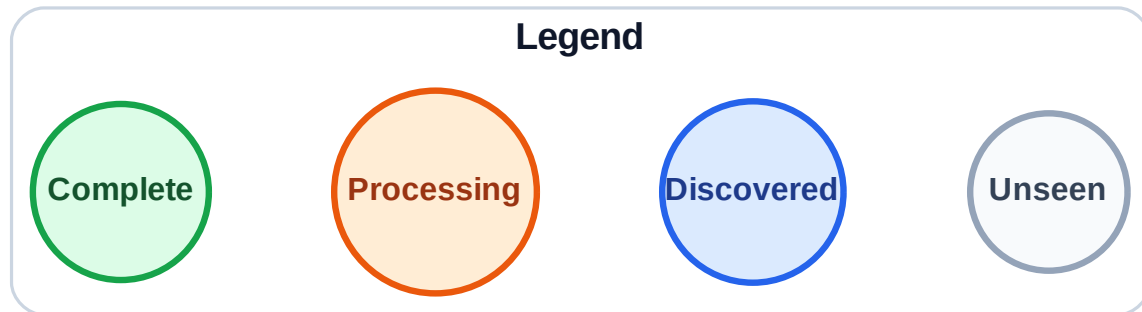
Unseen

Stack (top at right): M | E | D | C | L | K
Visited: A, B, F, G, H, I, N, O



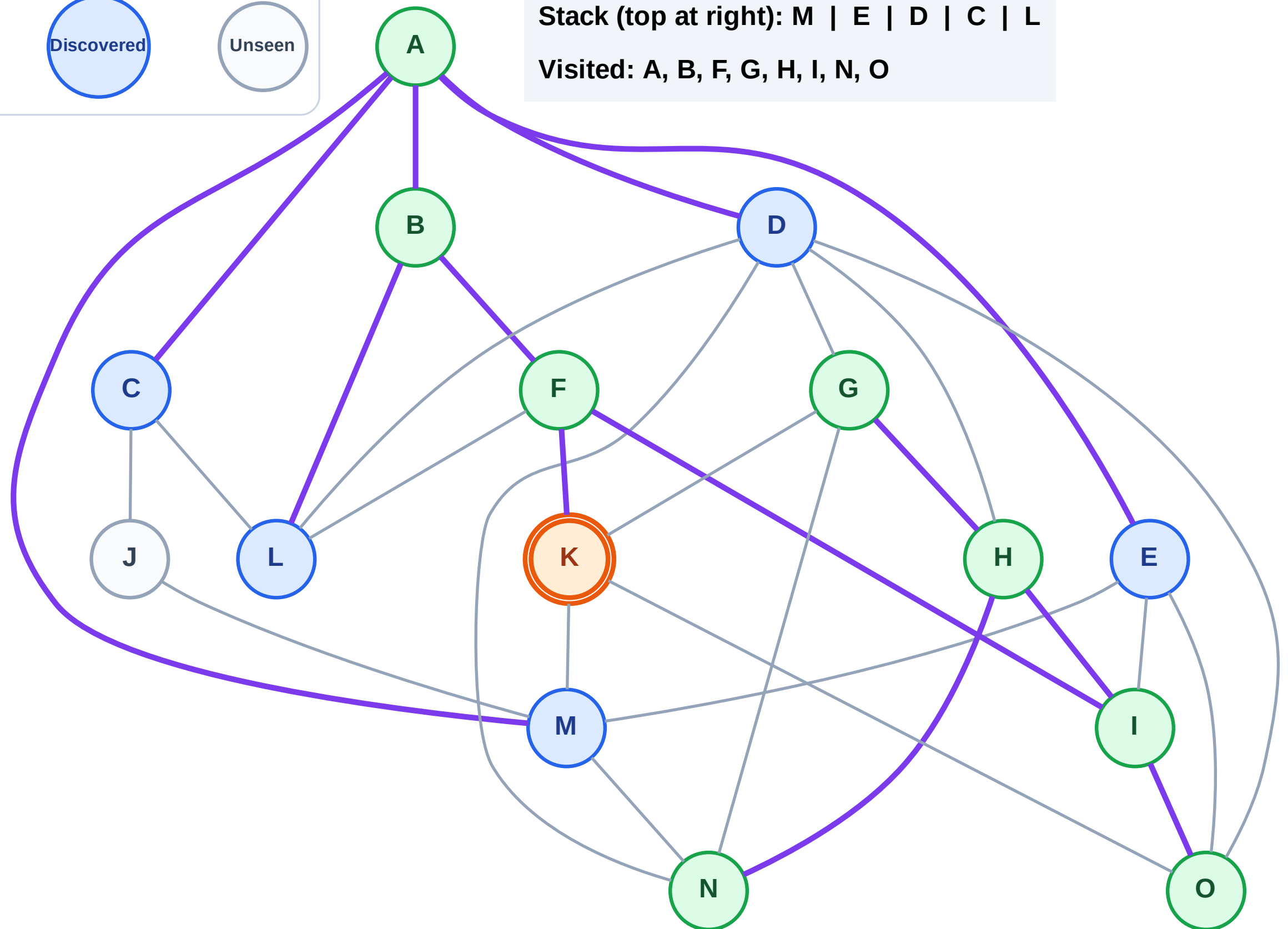
Step 18: Process node K

Step 18: Process node K



Stack (top at right): M | E | D | C | L

Visited: A, B, F, G, H, I, N, O



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (3 hops): A → B → F → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | E | D | C | L

Visited: A, B, F, G, H, I, K, N, O

